

The Big Pool Bangs Theory

Getting Rid of Dark Matter and Comic Inflation

Abstract. One of the two dark sectors can be dispensed with, the other cannot. In the framework presented here dark matter needs no new particle: it is the Planck-mass relics that survive the merger freeze-out in the innermost ring, which we call Kernel 0,0,0 (the origin) — the trace fraction $f = T_{\text{eq}}/T_{\text{conv}}$ left when relic mergers, which convert relic mass to light, stop keeping pace with the expansion — ordinary gravitating matter invisible to every non-gravitational detector, so the forty-year null of direct searches becomes a prediction (P5 (No Particle DM)) and a confirmed particle detection would falsify the core hypothesis; the $88\% \pm 4\%$ of the relic mass that the chaotic post-bounce rebound throws outward forms the dark rings outside Kernel 0,0,0, not the local dark-matter fraction (H8 (Precious Rings – Dante’s Purgatory)). Dark energy is *not* dispensed with: no arrangement of ejecta or of neighbouring universes can accelerate the expansion, and the acceleration is attributed to a constant tension of space whose value the framework does not derive. Cosmic inflation — “comic inflation” in our working vocabulary — is dispensed with as a *mechanism*: there is no inflaton field and no slow roll; a non-singular hard-sphere bounce replaces the initial singularity and evades the Borde–Guth–Vilenkin theorem. Its two classic jobs are, however, not yet done by anything else: the horizon and flatness problems remain open and the red tilt is fitted rather than derived (Sec. 5). With that scorecard stated, we explore a mechanical alternative to inflationary Λ CDM in which the universe begins as a jammed ensemble of Planck-mass relics ($\sim 10^{60}$ – 10^{61} per present Hubble volume), treated as effective hard spheres (an ansatz motivated by the Compton–Schwarzschild coincidence).¹ A loop-quantum-cosmology-type modified Friedmann equation, with critical density conjecturally identified with the hard-sphere jamming density, produces a non-singular bounce; the subsequent chaotic rebound ejects most of the ensemble. Toy N -body simulations with a turbulent kick distribution eject $88\% \pm 4\%$ of the mass, consistent with the observed dark-matter fraction for $\sigma_{\text{rel}} \approx 1$ — a regime a cascade nucleation model only marginally reaches — and show clean velocity sorting of the ejecta. Two geometric readings of that sorting are defined: Geometry A (homogeneous), in which the jammed patch is super-horizon and the sorting is velocity-space structure whose initial peculiar velocities must redshift into the Hubble flow; and Geometry B (finite centred cloud), in which the rebound is a single centred event and the radial ordering survives as a ballistic Hubble-like law. H8 (Precious Rings – Dante’s Purgatory) assigns them by location — Geometry A inside Kernel 0,0,0, Geometry B for the rings — so that the identical relics obey different mechanics in different rings set by one number, the kick relative to escape speed: the collisional Kernel 0,0,0 is the hot Big Bang we live in, the mantle a dark buffer, the free-streaming shell the dark rings. Geometry B still has *no bounce mechanism* for the system as a whole — a finite cloud of the relevant mass lies $\sim 10^{40}$ inside its own Schwarzschild radius and the homogeneous bounce equation offers it no exit — and its monopole distance–redshift relation with

¹Code, figures, build instructions and download links for all datasets used: <https://github.com/AIMFIRST-VN/big001>

the tension included has not been computed. The framework does *not* reproduce cosmic acceleration: coasting expansion is disfavored by Pantheon+ ($\Delta\chi^2 \approx 84$) and DES-SN5YR ($\Delta\chi^2 \approx 131$), a relativistic LTB lightcone through the ejecta decelerates ($q_0^{\text{eff}} = +0.16$), a toy effective $w(z)$ drifts in the wrong direction relative to the DES+DESI fit, and a kinetic-Sunyaev–Zel’dovich gate shows that no ejection profile can supply more than $\sim 1\%$ of the observed dark energy. We therefore adopt the Big Pool Bang framework *plus a constant tension of space* ($w = -1$) as the expansion history: the rebound sets the origin and initial velocity structure of the matter, the tension sets the acceleration, and the value of the tension is not derived. A predicted Cold Spot hot ring at $5\bar{\text{r}}\text{--}7\bar{\text{r}}$ is absent in Planck 2018 data. Because a neighbouring universe imprinting the last-scattering surface lies beyond our particle horizon, only an initial-condition (Sachs–Wolfe) imprint of its potential is causally admissible; for the Cold Spot this fixes a source mass $M_2 \approx 10^{48}$ kg ($+0.6/-0.4$ dex, an upper bound at fixed amplitude) 1–2.5 Gly beyond last scattering, whose induced velocity of the local volume is $\lesssim 1$ km s $^{-1}$. We organize the claims into eight hypotheses: the bounce-and-ejection core, H1 (Bounce – The Big Pool Bang); the rings, H8 (Precious Rings – Dante’s Purgatory), under which we live in Kernel 0,0,0 and the ejecta are the dark rings; the Cold Spot as a frozen vortex, H2 (Vortex – The Cold Eye of Sauron); a neighbouring Big Pool Bang in the Cold Spot direction, H3 (Neighbour – Big Pool Bang Multiverse 2); a population of such neighbours, H4 (Population – The Dark Comic Background); our own position inside Universe 1, H5 (Our Position – Where Is Nemo), which H8 (Precious Rings – Dante’s Purgatory) turns into a prediction — we sit in the core at or near its centre, so the structure-subtracted off-centre dipole should stay inside the 95th-percentile mock floor (≈ 2.5 Mpc at present), as it does, and a detected offset above it would falsify the placement H8 (Precious Rings – Dante’s Purgatory) assigns us; the location of the second Big Pool Bang, H6 (Neighbour Location); and the population model, H7 (Pool Model). Each carries explicit falsifiers collected in one table; we catalogue eleven predictions (P1 (Polarization Ring)–P11 (Cold Relics)) with the instruments required, and state the unsolved problems: causal structure of the bounce, relativistic formulation of ejection, the horizon and flatness problems, the expansion history, and the energy budget: a radiation era sourced by a single homogeneous population of relics leaves them relativistic at matter–radiation equality (hot dark matter), which H8 (Precious Rings – Dante’s Purgatory) resolves by location at the price of a merge-then-evaporate freeze-out calculation, baryogenesis, and the claim that Kernel 0,0,0 is homogeneous on our horizon scale (Secs. 3, 4).

1. Introduction: Three Questions

This paper is organized around three questions.

1. **Where are we?** If the universe began as a jammed ensemble of Planck-mass relics that rebounded and ejected most of its mass, the ejection was velocity-sorted (Sec. 3, Fig. 2). What that sorting means depends on the geometry, and this paper defines two readings (Sec. 3) and then assigns them by location (Sec. 4). In *Geometry A* (homogeneous) the jammed patch is super-horizon, ejection happens everywhere, and the sorting is velocity-space structure on an FLRW background: the ejection sets the matter’s initial peculiar-velocity distribution, which redshifts into the Hubble flow under the radiation era that relic conversion in Kernel 0,0,0 supplies (H8 (Precious Rings – Dante’s Purgatory)). In *Geometry B* (finite centred cloud) the rebound is a single centred event, genuine radial ordering survives — the fastest furthest — and the Hubble-like law is ballistic ($v = r/t$) plus the adopted tension — though Geometry B has, at present, no bounce mechanism for the system as a whole (Sec. 3). H8 (Precious Rings – Dante’s Purgatory) says both hold, in different rings: Geometry A inside Kernel 0,0,0, the innermost ring where the hot Big Bang happened and we live, and Geometry B for the dark rings around it. H5 (Our Position – Where Is Nemo; Sec. 11) then becomes a prediction

rather than a discriminator: we are in Kernel 0,0,0 at or near its centre, so a displaced-centre dipole above the mock floor, or a bulk flow that grows with depth, would falsify the placement H8 (Precious Rings – Dante’s Purgatory) assigns us. Our current answer: the displaced-centre test is null to within a few Mpc, as H8 (Precious Rings – Dante’s Purgatory) requires, the bulk flow at $\sim 430 \text{ km s}^{-1}$ is the only kinematic handle, and the hemispherical asymmetry is better assigned to an external neighbour than to our own offset.

2. **Where is the second Big Bang?** A neighbouring Big Pool Bang beyond our particle horizon can imprint the last-scattering surface only through the potential it contributed to the initial conditions — a Sachs–Wolfe spot with a low-multipole tail. The Cold Spot at $(l, b) = (203^\circ, -56^\circ)$ is the candidate, treated under H2 (Vortex – The Cold Eye of Sauron), H3 (Neighbour – Big Pool Bang Multiverse 2) and H6 (Neighbour Location) in Secs. 7–9: direction known to $\pm 3^\circ$, comoving distance 1–2.5 Gly beyond the last-scattering surface, mass $M_2 \approx 10^{48} \text{ kg}$ to within $+0.6 / -0.4$ dex, and a curl-mode test, P8 (Curl Test), pre-registered. The eight extreme spots of the Planck map are the candidate list for a population of such neighbours, H4 (Population – The Dark Comic Background) and H7 (Pool Model), treated in Sec. 10.
3. **How does the Big Pool Bang model work, and what does it explain?** A first-order vacuum crystallization jams into a hard-sphere state; a loop-quantum-cosmology-type bounce rebounds it; chaotic ejection sends $88\% \pm 4\%$ of the mass outward; this is H1 (Bounce – The Big Pool Bang), Secs. 2, 3 and 5. The identical relics then obey different mechanics in different rings, set by one number, the kick relative to escape speed; this is H8 (Precious Rings – Dante’s Purgatory), Sec. 4. The relics that fell back form Kernel 0,0,0, the innermost ring, where collisions grind relic mass into light until the mergers freeze out and leave a cold trace of survivors: that light is the hot Big Bang we live in, and those survivors are our dark matter, with no new particle. The 88% that escaped are the dark rings outside Kernel 0,0,0. In Geometry A, which holds inside Kernel 0,0,0, the ejection fixes the matter’s initial peculiar velocities on an assumed background; in Geometry B, which holds for the rings, it is a centred event whose radial ordering is the Hubble-like law itself. The rebound kinetics do *not* reproduce cosmic acceleration in either geometry, so the adopted expansion history is the Big Pool Bang framework plus a constant tension of space whose value is not derived (Sec. 6.2); the predicted Cold Spot thermal ring is absent, and the freeze-out calculation that fixes the survivor fraction has not been done (Sec. 13).

The paper answers the third question first — the bounce (Sec. 3), the rings (Sec. 4), the consistency checks (Sec. 5) and the expansion history (Sec. 6) — then treats the Cold Spot (Sec. 7), the neighbour mechanism (Sec. 8), its location (Sec. 9), the population (Sec. 10), and our own position (Sec. 11). Section 12 collects the eleven predictions and the single falsification table; Sec. 13 lists open problems.

1.1 Vacuum Crystallization and the Hard-Sphere State

Standard cosmology relies on scalar-field inflation to explain why the observable universe is flat, isotropic, and thermally uniform, at the cost of fine-tuning and of singularity theorems (Borde, Guth & Vilenkin 2003). The **Big Pool Bangs Theory** is a mechanical alternative grounded in first-order phase-transition dynamics. Space begins in a false vacuum with $\rho_{\text{vac}} \sim \rho_P$. Quantum fluctuations trigger a **vacuum crystallization** transition which nucleates at many points: bubbles of converted phase grow, collide, and merge, and the “bang” is their percolation into a single jammed state.

Timeline. Bubble walls propagate at $\leq c$. The comoving region that becomes one present-day Hubble volume spans, at the bounce density, $\sim 10^{20}$ causal (Planck-size) patches per dimension, so percolating it takes at least $10^{20} t_P = 5.4 \times 10^{-24}$ s divided by the number of nucleation sites per crossing length; any nucleation density sparse enough to give nontrivial merger statistics implies an assembly time many orders above t_P , whose consistency with trapped-surface avoidance (Sec. 3) is open.

Nucleation as the origin of the kick spectrum. The randomness of the merger history is inherited by the jammed core as internal turbulence, a candidate source for the stochastic kick spectrum that Sec. 3 otherwise requires as an input — provided nucleation-era velocity structure survives the collisional jammed epoch rather than thermalizing. A Poisson nucleation-and-merger simulation (vector-summed wall impulses) recovers a Maxwellian kick distribution (speed coefficient of variation 0.41–0.42), largely a central-limit consequence; the nontrivial statistic is the coherent-to-dispersive ratio σ_{rel} . Poisson nucleation is turbulence-dominated ($\sigma/\langle v_r \rangle \approx 5\text{--}8$) and cannot supply the coherent radial component the ejection fraction requires. A *cascade* model — a single seed bubble whose wall catalyzes further nucleation, halted by a jamming feedback at $\eta \approx 0.64$ — yields $\sigma_{\text{rel}} = 0.65\text{--}0.78$, robust across a $4\times$ range in trigger distance, against the $\approx 0.75\text{--}1.0$ window the dark-matter fraction requires: the cascade reaches that window only marginally. Three caveats gate the chain: the $88\% \pm 4\%$ figure comes from Maxwellian-ansatz N -body runs and the cascade’s heavier-tailed kick field has not been fed through the N -body pipeline; the model retains structural choices (trigger distance, nucleation lag, wall-area weighting, the arrest $\eta = 0.64$); and the wall-momentum efficiency is assumed universal, whereas in first-order-transition physics it is model-dependent and typically small (Espinosa et al. 2010). Bubble collisions in any first-order transition also source a stochastic gravitational-wave background peaked at the transition scale; this is a generic prediction of H1 (Bounce – The Big Pool Bang), listed as P3 (Transition GW Background), not of any neighbour.

Because quantum geometry prohibits localized infinite densities, the energy crystallizes into irreducible **Planck-mass relics** ($m_P \approx 21.8 \mu\text{g}$). We count them *per present-day Hubble volume*: $N \approx 10^{60}\text{--}10^{61}$ relics, of mass $M_1 = Nm_P \approx 2.2 \times 10^{52}$ kg at the lower end (order of magnitude; the matter mass of the *observable* universe is $\sim 1.5 \times 10^{53}$ kg). These are the relic number and mass per Hubble volume today, not totals: in Geometry A (Sec. 3) the jammed patch is super-horizon, so its total number and mass are unbounded by observation; in Geometry B they are the mass of a finite cloud, at least the $\sim 1.5 \times 10^{53}$ kg of the observable universe. The relics saturate local volume up to the random close-packing limit $\eta_{\text{rcp}} \approx 0.64$, and with $\rho_P = m_P/\ell_P^3 = 5.2 \times 10^{96}$ kg m $^{-3}$ the jamming density is $\rho_c = \eta_{\text{rcp}}\rho_P$. Compressed to ρ_c , one Hubble volume’s worth of relics occupies a region $\sim 10^{-15}$ m $\approx 10^{20} \ell_P$ across — but this must not be read as the size of an isolated object: the Schwarzschild radius of 2.2×10^{52} kg is 3.3×10^{25} m, forty orders larger, so a finite ball of this mass and density would be a black hole, not a bouncing patch. The bounce of Sec. 3 is meaningful only if the jammed patch is the whole (or a super-horizon) universe described by a homogeneous FLRW-type metric, for which no exterior Schwarzschild geometry exists (Geometry A of Sec. 3); the finite-cloud reading (Geometry B) must confront this trapped-surface problem directly. We therefore use ρ_c , not a jamming radius, as the physical parameter.

2. Planck-Mass Relics as Effective Hard Spheres

For a particle of mass m the reduced Compton wavelength $\lambda_C = \hbar/mc$ and the Schwarzschild radius $R_s = 2Gm/c^2$ coincide at $m \approx m_P$, where both equal $\sim \ell_P = 1.616 \times 10^{-35}$ m. This is the regime in which neither quantum field theory nor classical general relativity is separately valid. **We adopt, as a modeling ansatz rather than a derivation, that Planck relics possess a**

minimum spatial extent $\sim \ell_P$ **that no interaction can compress**, and treat them as effective incompressible hard spheres. The ansatz is judged by the phenomenology it generates (Secs. 3–11); its microphysical justification is an open problem (Sec. 13).

3. H1 (Bounce – The Big Pool Bang): The Kinetic Bounce and Chaotic Ejection

Why hard-sphere pressure cannot bounce the collapse. Inside a homogeneous fluid the dynamics are governed by the Raychaudhuri equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3P}{c^2} \right).$$

Near random close packing the hard-sphere kinetic pressure diverges (free-volume equation of state, $P \propto nk_B T [1 - (\eta/\eta_{\text{rcp}})^{1/3}]^{-1}$), but in general relativity positive pressure *gravitates*: no classical equation of state with $P > 0$ can bounce an FLRW collapse. The rebound must be sourced by quantum-geometric corrections that dominate at the jamming density $\rho_c = \eta_{\text{rcp}} \rho_P$. The minimal object with the required property is the loop-quantum-cosmology (LQC) modified Friedmann equation,

$$H^2 = \frac{8\pi G}{3} \rho \left(1 - \frac{\rho}{\rho_c} \right), \quad \rho_c = \eta_{\text{rcp}} \rho_P,$$

which encodes a maximum attainable density covariantly: $H \rightarrow 0$ as $\rho \rightarrow \rho_c$ and the effective Raychaudhuri equation flips sign. We propose the jamming limit of the relic gas as the physical origin of the ρ_c that LQC introduces axiomatically, noting that LQC’s own Immirzi-fixed value is $\rho_c \approx 0.41 \rho_P$ versus $0.64 \rho_P$ here — order-of-magnitude agreement, a coincidence rather than a derivation. A numerical integration (Fig. 1) confirms that the classical hard-sphere fluid overshoots the jamming density by $> 10^4$ while the modified equation bounces at $\rho = \rho_c$; the bounce is built into the equation, not independently confirmed by it. Whether it completes before a trapped surface forms around the still-homogeneous patch is an assumption (Sec. 13).

Regime disclaimer. Everything that follows in this section is a **Newtonian analogy**: the real dynamics occur where escape velocities formally exceed c by ~ 20 orders of magnitude and only a general-relativistic treatment is meaningful. We use the Newtonian model because violent-relaxation statistics (which fraction unbinds, how ejecta sort by speed) are plausibly governed by dimensionless energy ratios that survive the translation; whether they do is the central open problem of Sec. 13.

The slingshot phase. Picture the jammed core as 10^{60} billiard balls packed to the jamming limit; when the bounce reverses the collapse, the rebound plays the role of the opening break. At jamming density the relic gas is collisional, so the slingshot phase begins only after expansion dilutes the ensemble below the contact threshold and relics decouple onto ballistic trajectories. From then on the system is a dense self-gravitating N -body ensemble in which three-body encounters dominate relaxation — the “core collapse and halo ejection” of globular-cluster dynamics, which sheds most of a system’s mass to reach a bound remnant. Applied to the birth of the universe it sorts the mass into two components:

- **The ejected halo ($\approx 88\%$):** relics kicked above escape velocity, free-streaming and rarely colliding again — the dark rings outside Kernel 0,0,0 under H8 (Precious Rings – Dante’s Purgatory), ordinary gravitating mass requiring no new particle (but see the temperature caveat below).

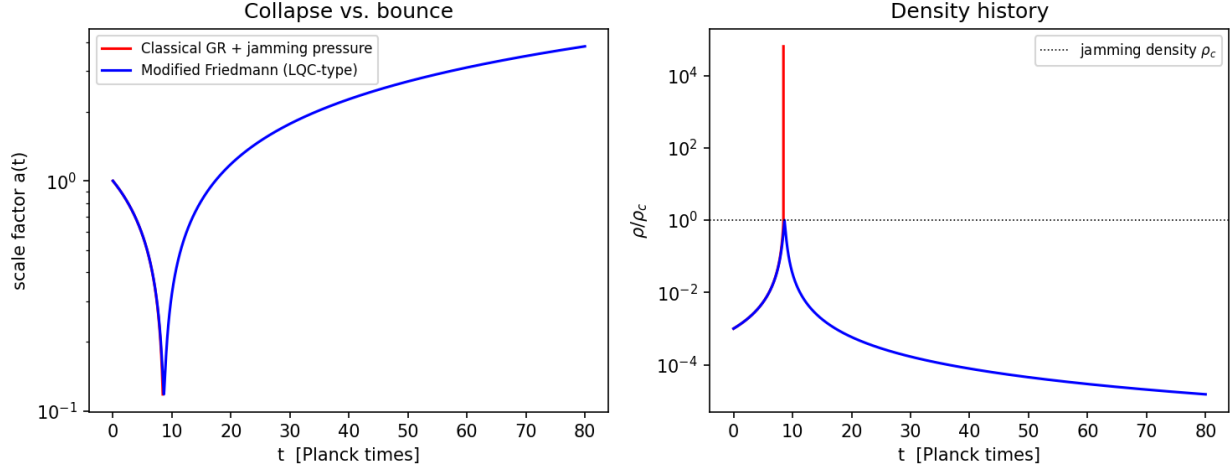


Figure 1: Fig. 1 — Toy FLRW integration. Left: scale factor. Right: density history. Classical GR with jamming pressure (red) overshoots the jamming density by $> 10^4$ (collapse); the modified Friedmann equation (blue) bounces at ρ_c .

- **The trapped core ($\approx 12\%$), Kernel 0,0,0:** relics that fall back and grind against each other in a turbulent core, proposed as the origin of the photon–baryon plasma and, through the survivors of that grinding, of the local dark matter (Sec. 4). **Caveat (essential):** this requires a conversion channel from Planck relics to Standard-Model plasma — including baryogenesis and the photon-to-baryon ratio — while ejected relics remain stable for 13.8 Gyr. *Conjectured channel (merge-then-evaporate).* A single Planck-mass relic is precisely the object usually taken to be the stable end state of black-hole evaporation, so the stability of isolated ejecta is the standard assumption rather than an extra one. In the trapped core, however, relics collide and merge; a merger product of mass $2m_P$ is above the remnant scale and evaporates back to one relic in a few Planck times, radiating $\sim m_P$ into a thermal bath of Standard-Model quanta per merger. Repeated in a dense, collisional core this converts relic mass into plasma at a rate set by the merger rate, while the dilute ejecta, whose collision rate is negligible, stay as relics. This is a conjecture in the semiclassical regime where Hawking’s calculation is least trustworthy, and it produces a CP-symmetric bath, so baryogenesis is not supplied by it; both remain open (Sec. 13). Absent such a channel (e.g. a threshold collision energy the ejecta never reach), the trapped fraction cannot be identified with the baryon budget.

Ejecta temperature and the bounce-to-equality energy budget. The ejecta leave relativistically, so they must be cold ($p/m_{PC} \lesssim 10^{-3}$) by matter–radiation equality or they are hot dark matter. Free-streaming momenta redshift as $p \propto 1/a$, so the question is how much expansion separates the bounce from equality, and that is fixed by the energy budget: $a_{\text{eq}}/a_{\text{bounce}}$ equals the ratio of radiation-like to rest-mass energy at the bounce. We evaluated this ratio for every source of radiation the framework offers (`energy_budget.py`). (i) If the radiation is the relics’ own kinetic energy, then radiation domination ends exactly when the relics stop being relativistic, because it is the same energy redshifting: for any ejection Lorentz factor from 2 to 10^{20} , with or without the trapped core converting all of its rest mass, the ejecta reach equality with $p/m_{PC} = 1.6\text{--}1.9$. They are hot dark matter, independent of γ . (ii) The trapped core cannot rescue this: the trapped relics

are the slow ones by velocity sorting, so they carry less energy than the ejecta; cold ejecta would require the trapped population to hold $\gtrsim 100$ times the ejecta’s energy per particle. (iii) Formally, a radiation component $R_0 \gg \gamma$ (in units of the relic rest energy) that is *not* sourced by the relics would give $p/m_{pc} = \gamma/R_0$ at equality — but it would also make the relics a trace $1/R_0$ of the energy at the bounce, with packing fraction $\sim 0.64/R_0 \lesssim 10^{-3}$, so that nothing is jammed and the bounce is a bounce of a radiation fluid; and the framework admits no such component, since it contains no residual vacuum and no reheating stage. Within a single homogeneous history, then, the jamming bounce and cold Planck-relic dark matter do not fit together. H8 (Precious Rings – Dante’s Purgatory; Sec. 4) resolves this by location: the relics that fell back and the relics that escaped have different histories, and the budget that matters for the CMB is that of the trapped core alone.

Ejection fraction (toy N -body). A softened $N = 1000$ equal-mass simulation of a cold uniform sphere given a radial kick $v = f v_{\text{esc}}$ (with v_{esc} the surface escape speed) has a steep ejection threshold: $f = 0.6 \rightarrow 3\%$, $0.8 \rightarrow 41\%$, $0.9 \rightarrow 98\%$, $f \geq 1 \rightarrow 100\%$. A monolithic kick reproduces the dark-matter fraction only in the narrow window $f \approx 0.85\text{--}0.9$, and the surface-escape-speed kick ($f = 1$) over-ejects. A Maxwellian turbulent component of dispersion $\sigma_{\text{rel}} \bar{f} v_{\text{esc}}$ superimposed on a mean rebound $\bar{f} v_{\text{esc}}$ broadens the transition: at $\bar{f} = 0.7$ the ejected fraction rises from 22% ($\sigma_{\text{rel}} = 0.5$) through $\approx 88\%$ (1.0) to 98% (1.5). Over a 15-seed ensemble at $(\bar{f}, \sigma_{\text{rel}}) = (0.7, 1.0)$ the ejected fraction is $88\% \pm 4\%$, consistent with the observed dark-matter mass fraction, at a σ_{rel} the cascade model of Sec. 1 only marginally reaches; $(\bar{f}, \sigma_{\text{rel}})$ remain inputs to be derived from the bounce turbulence spectrum.

Ejecta kinematic structure (15 seeds, 13{,}140 ejecta). Ejection is extended (median ejection time ~ 1 dynamical time; 10% of ejecta leave after 11), with clean monotonic velocity sorting: median speeds fall from $0.71 v_{\text{esc}}$ for the earliest ejecta to $0.28 v_{\text{esc}}$ for the latest (Fig. 2), so radial position maps onto ejection epoch. The tangential structure shows no correlated anisotropy: the per-sky ejection dipole (0.064 ± 0.024) matches shot noise (0.053 ± 0.024), and early-vs-late shell axes align only weakly ($\langle \cos \theta \rangle = 0.37 \pm 0.49$). The isotropic cascade therefore supplies no preferred axis; if one exists it must be seeded externally (Sec. 8).

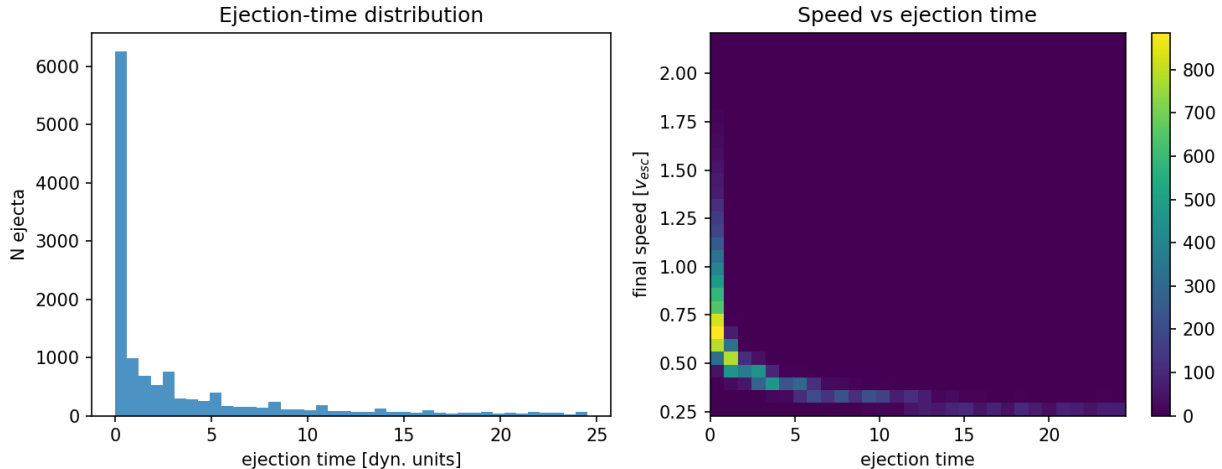


Figure 2: Fig. 2 — Ejecta kinematics over 15 seeds: ejection-time distribution (left) and speed vs ejection time (right), showing the velocity-sorted shell structure.

From the N -body cloud to the universe: two geometries. The N -body ejecta cloud is a

sub-horizon toy of the ejection *mechanism*. How it maps onto the universe is not settled by the toy, and this paper carries two geometric hypotheses in parallel.

- **Geometry A (homogeneous).** The jammed patch is the whole (super-horizon) universe (Sec. 1), there is no exterior, and ejection happens everywhere: every comoving region is simultaneously “trapped core” and “halo”. The radial shells of Fig. 2 translate into *local velocity-space structure* on an assumed FLRW background, not into a centred cloud with an edge. What survives is the statistical content (ejected fraction, speed distribution, isotropy); what does not is the geometric centre. The ejection sets the matter’s *initial peculiar-velocity distribution*, which redshifts into the Hubble flow under the radiation era that relic conversion in Kernel 0,0,0 supplies (Sec. 4). This geometry avoids the trapped-surface problem of Sec. 1.1 and predicts a null for H5 (Our Position – Where Is Nemo).
- **Geometry B (finite centred cloud).** The rebound is a single centred event and genuine radial velocity sorting survives: the fastest furthest. The Hubble-like law is ballistic, $v = r/t$, plus the adopted tension of Sec. 6.2. Two burdens must be stated plainly. First, **Geometry B currently has no bounce mechanism.** A finite cloud of the relevant mass ($M_1 = 2.2 \times 10^{52}$ kg per Hubble volume, and at least the 1.5×10^{53} kg of the observable universe) compressed to ρ_c is $\sim 10^{-15}$ m across against a Schwarzschild radius of $\sim 10^{25}$ m — some 10^{40} inside its own horizon — and the homogeneous LQC equation above, written for a metric with no exterior, offers such a cloud no exit: it is a trapped region, and nothing in this paper takes it out again. Geometry B is retained not because a bounce has been shown to occur in it but because it is the geometry that H5 (Our Position – Where Is Nemo) can test kinematically. Second, its principal observational burden is the *monopole* distance–redshift relation, not the dipole: a few-Mpc offset in a Gpc-scale cloud is trivially isotropic, so supernova isotropy is not the issue; the issue is that the ballistic-only LTB lightcone through the ejecta (Appendix C) decelerates ($q_0^{\text{eff}} = +0.16, \sim 0.35$ mag from Λ CDM by $z \sim 5$), and the LTB-plus-tension lightcone that Geometry B actually requires has not been computed (Sec. 6.2). What H5 (Our Position – Where Is Nemo) directly tests is our offset from the centre and radial ordering in the bulk flow.

H8 (Precious Rings – Dante’s Purgatory; Sec. 4) assigns the two geometries by location: Geometry A inside Kernel 0,0,0, Geometry B for the rings. The finite-cloud burdens above are then burdens on the system of rings as a whole, not on the region we observe, and H5 (Our Position – Where Is Nemo; Sec. 11) becomes a prediction: we are in Kernel 0,0,0 at or near its centre, so a structure-subtracted off-centre dipole above the 95th-percentile mock floor with a direction stable across shells, or a bulk flow that grows with depth, would falsify the placement H8 (Precious Rings – Dante’s Purgatory) assigns us, while an offset consistent with the mock distribution together with a flow that decays with depth is what it requires.

4. H8 (Precious Rings – Dante’s Purgatory): Kernel 0,0,0 and the Dark Rings

Motivation. The bounce-to-equality budget of Sec. 3 shows that a single homogeneous bounce cannot be both of the things the framework asks of it: a jammed-relic rebound whose ejection fraction is the dark-matter budget, and the source of a radiation era that leaves cold relics at matter–radiation equality. If every relic shares one history, the ejecta are hot at equality for any ejection Lorentz factor. H8 (Precious Rings – Dante’s Purgatory) resolves the conflict by *location*.

The relics are identical, but they do not share one history: the kick each receives relative to the local escape speed, $f = v/v_{\text{esc}}$ — the one number the rebound assigns to every relic (Sec. 3, Fig. 2) — sorts them into concentric rings, and the mechanics differ from ring to ring because the density, the collision rate and the binding differ. Dante’s Purgatory is the picture: one mountain, one kind of soul, and a different law on every terrace. The innermost ring, which we call Kernel 0,0,0, is where the relics that fell back collide, and it is where we live. Table 7 is the map.

Table 7. H8 (Precious Rings – Dante’s Purgatory): the rings, their mechanics, and our handle on each.

Ring	Who is there	Mechanics	State	Becomes	Our handle
Kernel 0,0,0 (the trapped core)	Relics that fell back (f well below 1)	Collisional: mergers, evaporation, thermalization, radiation-fluid dynamics	Light plus a trace of surviving relics	The hot Big Bang we live in: CMB, helium, baryons, and our cold dark matter via merger freeze-out	Everything we observe
Mantle	Relics that rose and fell back late, or are marginally bound ($f \approx 1$)	Gravitational, weakly collisional, mixed by many orbits	Cold relic gas, no light	A dark buffer with no memory of the initial geometry	Sets whether Kernel 0,0,0 has an edge, and how an H5 (Our Position – Where Is Nemo) offset would show
Shell	The fast ejecta (the 88%, $f > 1$)	Collisionless free streaming, fastest furthest	Cold, dark, velocity-sorted	The dark rings around us; the only ring that keeps the initial geometry and spin	Beyond our horizon except through initial-condition imprints

Ring	Who is there	Mechanics	State	Becomes	Our handle
Boundary and beyond	Walls to neighbouring bangs	Vacuum- bubble physics (Coleman & De Luccia 1980)	Not matter	Cold Spot imprint, low-multipole tail	H3 (Neighbour – Big Pool Bang Multiverse 2), H6 (Neighbour Location), P8 (Curl Test), P9 (Low- ℓ Tail)

We live in Kernel 0,0,0. The hot Big Bang is Kernel 0,0,0. Once the budget is written for Kernel 0,0,0 alone (`energy_budget.py`, case 4), the conflict of Sec. 3 disappears, because the radiation and the dark matter of Kernel 0,0,0 are no longer required to be the same relics at the same speed. The grinding in Kernel 0,0,0 converts a fraction $1 - f$ of its relics to light by the merge-then-evaporate channel of Sec. 3: a merger product above the remnant mass evaporates, radiating $\sim m_P$ of Standard-Model quanta. Conversion is self-limiting. As Kernel 0,0,0 expands and its density drops, the merger rate falls below the expansion rate and mergers *freeze out*, exactly as a WIMP’s annihilations do; the survivors, a fraction f , are the local cold dark matter. Consistency with the observed matter–radiation equality, $T_{\text{eq}} \approx 0.8$ eV, fixes the survivor fraction in terms of the temperature T_{conv} at which conversion stops: the radiation-to-rest-mass ratio at freeze-out is $R = (1 - f)/f$, equality occurs when it has redshifted to unity, $a_{\text{eq}}/a_{\text{conv}} = R = T_{\text{conv}}/T_{\text{eq}}$, so

$$f = \frac{T_{\text{eq}}}{T_{\text{conv}}} : \quad f = 7 \times 10^{-29} \text{ (Planck-scale conversion)}, 8 \times 10^{-26} (10^{16} \text{ GeV}), 8 \times 10^{-20} (10^{10} \text{ GeV}), 8 \times 10^{-13} (\text{TeV})$$

The survivors are cold in every case: a relic that leaves the collisional phase with momentum γm_{PC} has $p/m_{PC} = \gamma/R \approx \gamma f$ at equality, from 10^{-28} (Planck-scale conversion) to 10^{-6} (MeV) for $\gamma \sim 1$, and still below 10^{-3} at $\gamma = 10^3$ for every conversion temperature listed. One relation, one new number to fix (T_{conv} , set by the merger cross-section and the remnant lifetime), and nothing else: the light comes from the relics. We state explicitly, because the subtitle promises it, that H8 (Precious Rings – Dante’s Purgatory) contains **no inflation-like phase of any kind** — no residual vacuum, no vacuum-dominated stretch, no reheating, no e-folds. The radiation era is sourced by merge-then-evaporate conversion in Kernel 0,0,0, the rings resolve the budget without any inflationary or reheating stage, and the crystallization’s latent heat plays no role in the accounting.

The 88% reassigned. The $88\% \pm 4\%$ ejected fraction of Sec. 3 is the mass of the *dark rings around Kernel 0,0,0* — the shell — and not the local dark-matter fraction. Inside Kernel 0,0,0 the dark-matter-to-baryon ratio, 5.4, is set by the freeze-out survivor fraction f together with baryogenesis in the bath, which the merge-then-evaporate channel does not supply and which remains open (Sec. 13). The abstract and Sec. 3 are worded accordingly: dark matter is the Planck-relic survivors of the freeze-out in Kernel 0,0,0; the ejecta form the rings outside it. P5 (No Particle DM) is unchanged — the survivors are the same objects, invisible to every non-gravitational detector.

The coherence gradient, reframed. Appendix A records, as an untested hypothesis, that only the earliest and fastest ejecta preserve geometric correlations while everything slower is randomized by violent relaxation. H8 (Precious Rings – Dante’s Purgatory) makes that its statement: free streaming preserves the initial geometry and spin in the shell; collisions in Kernel 0,0,0 and orbit mixing in the mantle erase them. The shell is the only ring that remembers how the bang began, and it is the ring we cannot see.

Geometry A and Geometry B, resolved by location. The two geometries of Sec. 3 are not alternatives once the rings are distinguished. Kernel 0,0,0 is Geometry A on our horizon scale: homogeneous, without a centre we can measure, with the collisional rebound and the hot Big Bang taking place everywhere within it. The whole system of rings is Geometry B: a single centred event whose radial ordering survives in the mantle and shell. H8 (Precious Rings – Dante’s Purgatory) therefore replaces the A/B hedge with a definite claim — Geometry A inside Kernel 0,0,0, Geometry B for the rings — and Secs. 3, 6.2 and 11 and the Epilogue are worded accordingly. For H5 (Our Position – Where Is Nemo) the consequence is a prediction rather than a discrimination: we are in Kernel 0,0,0, at or near its centre, and the displaced-centre test should be null, as it is (Sec. 11). The H5 (Our Position – Where Is Nemo) support criterion — an off-centre dipole above the mock floor with a direction stable across shells — would therefore falsify the placement H8 (Precious Rings – Dante’s Purgatory) assigns us.

Observers live in kernels. Only Kernel 0,0,0 has light. An observer in the shell would see no CMB, no stars, and no thermal history at all — Dante inverted: we are in the innermost ring, the only one with light, and that is why we are here to ask.

Falsifiers. H8 (Precious Rings – Dante’s Purgatory) falls to any of: (i) a detected off-centre dipole above the mock floor, with a direction stable across shells — we would not be at the centre of Kernel 0,0,0; (ii) any measured thermal velocity dispersion of dark matter above γ/R — a warm-dark-matter detection at the keV scale would falsify it, whereas the existing Lyman- α limits, $m_{\text{WDM}} \gtrsim \text{few keV}$, H8 (Precious Rings – Dante’s Purgatory) passes trivially; (iii) failure of the merge-then-evaporate freeze-out, once computed (merger rate, remnant stability), to reproduce $f = T_{\text{eq}}/T_{\text{conv}}$; (iv) any evidence that the CMB region has an edge within the horizon. These are collected in Table 2. The positive prediction is P11 (Cold Relics): the dark-matter thermal velocity dispersion at equality is below $10^{-6}c$ for any conversion temperature above the MeV scale, so the matter power spectrum has no free-streaming cutoff down to the smallest scales — no warm-dark-matter suppression — testable with the Lyman- α forest, strong-lens substructure and 21-cm tomography; consistent at present, not yet discriminating (Table 1).

5. Consistency Checks

- **Horizon problem — open.** At the bounce the causal patch is of Planck size, and the region that becomes one present-day Hubble volume comprises $\sim 10^{60}$ such patches (of order one per relic). Nothing in the bounce equilibrates them within a few Planck times, so homogeneity across causal patches is an assumed initial condition — in Geometry A over a super-horizon, unbounded patch; in Geometry B over the finite cloud. A long contracting phase before the bounce could in principle enlarge the causal patch, but the collapse is not modelled here.
- **Flatness — not addressed.** In the turbulent regime the simulations favor ($\bar{f} = 0.7$, $\sigma_{\text{rel}} = 1.0$) the mean kinetic energy is $\langle v^2 \rangle = \bar{f}^2(1 + \sigma_{\text{rel}}^2)v_{\text{esc}}^2 \approx v_{\text{esc}}^2$, but this does *not* make the total energy vanish. For a uniform sphere $U = -3GM^2/5R$ while the surface escape speed gives $v_{\text{esc}}^2 = 2GM/R$, so $\langle v^2 \rangle = v_{\text{esc}}^2$ means $K \approx 1.7|U|$

(or $K \approx 2|U|$ if the kick is scaled to the local escape speed); $E = 0$ would require $\bar{f}^2(1 + \sigma_{\text{rel}}^2) \approx 0.5\text{--}0.6$ (of order 0.3–0.6 depending on the escape-speed convention and kick weighting). The simulated systems are therefore unbound, $E > 0$, which is consistent with the 88% ejected fraction and with the surface-escape-speed kick $f = 1$ ejecting 100%. The energy budget of the rebound says nothing about spatial flatness; flatness is not addressed by this framework.

Remark on the spectral tilt. One may parametrize the scalar index by a turbulent dissipation ratio ν_t (eddy turnover time to expansion time) as $n_s - 1 = -\nu_t/3$; Planck’s $n_s = 0.965 \pm 0.004$ then gives $\nu_t \approx 0.105$. Both the relation and the value are fitted inputs, and showing that a Kolmogorov cascade ($E(k) \propto k^{-5/3}$) can produce a nearly scale-invariant curvature spectrum is an open problem (Sec. 13).

6. Expansion History: Why the Big Pool Bang Framework Needs a Constant Tension

The rebound kinetics alone imply a coasting expansion ($a \propto t$), and this is excluded. Against 1371 Hubble-flow SNe Ia from Pantheon+ (Scolnic et al. 2022; Brout et al. 2022; diagonal errors, offset marginalized), coasting yields $\chi^2 = 674$ vs 590 for flat Λ CDM ($\Delta\chi^2 \approx 84$), though far better than Einstein–de Sitter ($\chi^2 = 1170$; Fig. 3); on DES-SN5YR (DES Collaboration 2024; Dovekie recalibration, Popovic et al. 2025; 1820 SNe, full covariance) the figures are 1632, 1763 ($\Delta\chi^2 \approx 131$), and 2566. Whether the *distribution* of the ejecta could mimic acceleration for an interior observer is tested in Appendix C: a relativistic Lemaître–Tolman–Bondi lightcone through the measured ejecta shells decelerates ($q_0^{\text{eff}} = +0.16$); no smooth, bimodal, or lumpy ejection profile accelerates; a blast-wave cavity is the giant-void geometry; and the kinetic Sunyaev–Zel’dovich limit on coherent cluster flows ($v \lesssim 300 \text{ km s}^{-1}$ at Gpc distances) caps any cavity’s contribution at $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. **No ejection profile of any morphology supplies more than $\sim 1\%$ of the observed dark energy.**

6.1 The $w(z)$ drift

Integrating the measured ejecta speed distribution as cold spherical shells decelerating in their enclosed mass, and computing the equation of state an interior observer would fit, gives w_{eff} drifting from -0.29 at $z = 2$ to -0.31 today (Fig. 4): a few-percent drift near the coasting value $-1/3$. Because this apparent equation of state is set by the evolving ratio of velocity-dispersion energy to clustered rest-mass energy, it cannot be constant. The DES-SN5YR + DESI DR2 BAO (+CMB) $w_0 w_a$ CDM fit prefers evolving w at $\sim 3\sigma$ ($w_0 \approx -0.8$, $w_a \approx -0.7$; DES Collaboration 2024; DESI Collaboration 2025), with w becoming *less* negative with time; the toy shells drift the *opposite* way. The remap is crude (91% of the ejecta re-bind in a static enclosed-mass model, so the flow is built from the fastest 9%), but as it stands the kinetic sector shares with the data only the qualitative statement that w evolves, and predicts the wrong sign of the drift.

6.2 The adopted expansion history: the Big Pool Bang framework plus a constant tension

The failures above are failures of one idea only: that the *kinetics* of the rebound could substitute for dark energy. They do not touch the bounce, the ejection, or the dark-matter fraction. We therefore separate the two sectors. The Big Pool Bang framework is responsible for the origin and velocity structure of the matter; the acceleration is attributed to a constant tension of space, $\rho_\Lambda c^2 = -p_\Lambda$,

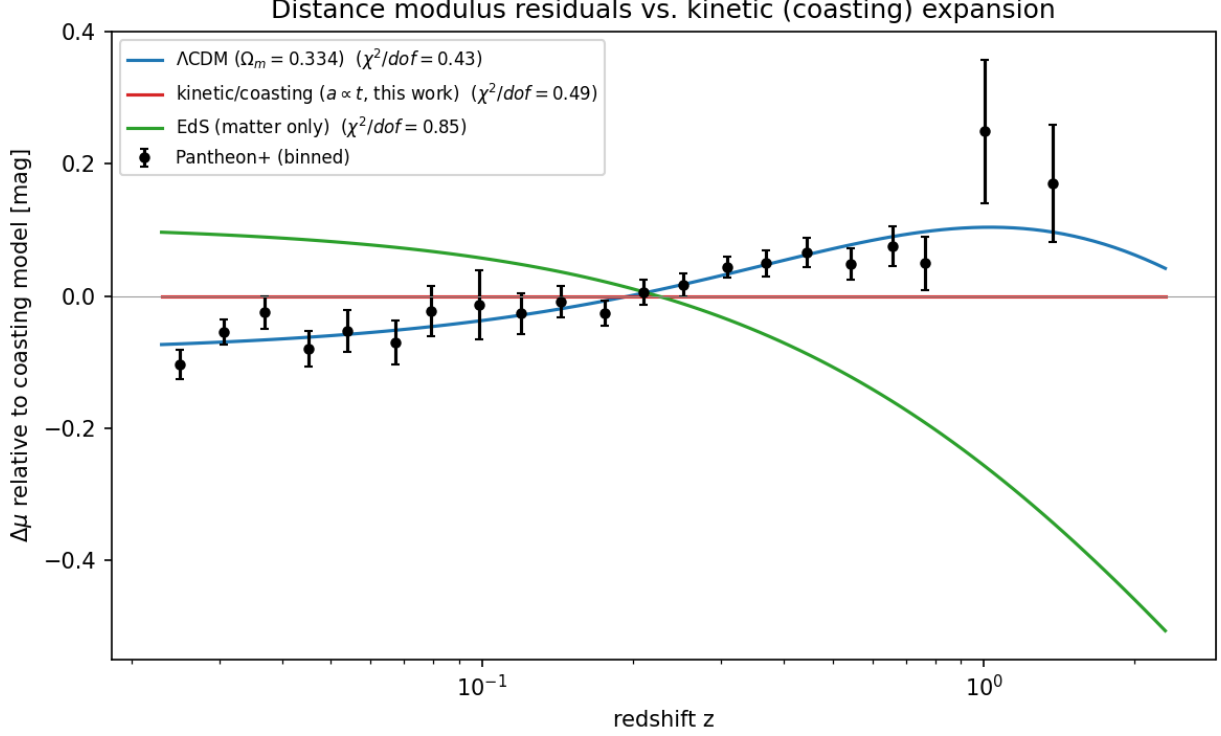


Figure 3: Fig. 3 — Pantheon+ binned distance-modulus residuals relative to the coasting model.

which we adopt without claiming to derive it.

Why a tension. For a comoving separation $d = a\chi$, $\ddot{d} = (\ddot{a}/a)d$ with $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3p/c^2)$: acceleration is proportional to distance, with a coefficient set locally by density and pressure, and is repulsive only for $\rho + 3p/c^2 < 0$. If space stores an energy density $u(a)$ when stretched, thermodynamics ($p = -\partial(uV)/\partial V$) fixes its pressure: for $u \propto a^{-n}$, $w = n/3 - 1$. Matter ($n = 3$) attracts; strings or curvature ($n = 2$, $w = -1/3$) coast; a domain-wall *network* ($n = 1$, $w = -2/3$) repels; a constant tension ($n = 0$, $w = -1$) repels; an energy growing as a^2 ($w = -5/3$) is phantom. The wall-network value applies to many walls per Hubble volume (Bucher & Spergel 1999); a single wall at ~ 46 Gly contributes no such term. Observationally, Planck+BAO+SN constrain a constant equation of state to $w = -1.03 \pm 0.03$ (Planck Collaboration 2020 VI), while the DES+DESI $w_0 w_a$ fit prefers evolving w at $\sim 3\sigma$ (Sec. 6.1). We adopt constant $w = -1$ as the minimal assumption consistent with the first, noting that the evolving- w hint, if confirmed, is not something the kinetic sector can explain: its toy drift has the wrong sign. Neighbouring Big Pool Bangs cannot contribute: a single neighbour supplies a tidal shear and a negligible induced velocity (Sec. 9), a symmetric population supplies nothing by the shell theorem (Appendix B), and neither is a monopole. Today the tension term gives $\ddot{a}/a = +3.4 \times 10^{-36} \text{ s}^{-2}$ against $-0.7 \times 10^{-36} \text{ s}^{-2}$ from matter.

Vacuum energies of neighbours do not sum. Vacuum energy acts only where it resides: the expansion rate of our region is fixed by the tension inside it, so the tensions of any number of neighbouring bangs contribute nothing here, for the same locality reason that their masses contribute no monopole (Appendix B). If all bangs crystallized into the same vacuum, the tension is one universal constant whose value the model does not predict; if into different vacua, the differences reside in the walls between them, and a higher-tension neighbour drives its wall outward as a bubble collision.

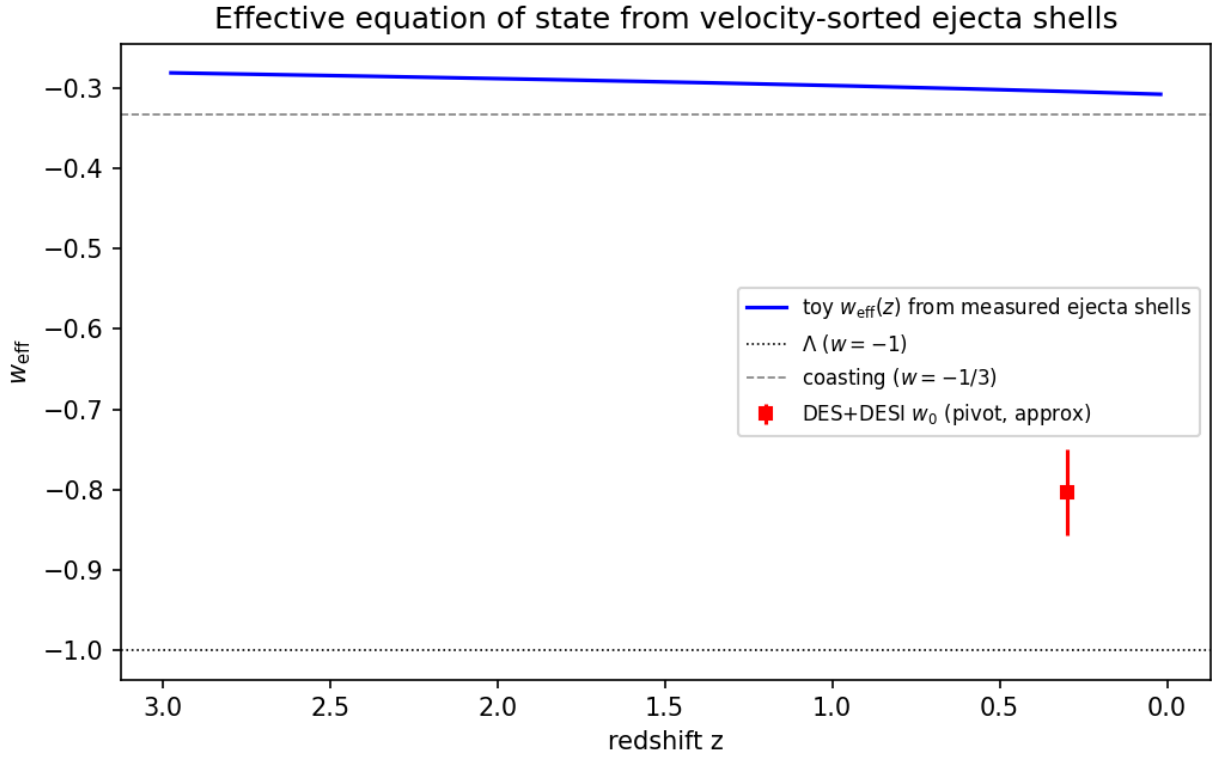


Figure 4: Fig. 4 — Toy effective equation of state from velocity-sorted ejecta shells: a few-percent drift near $w = -1/3$, in the opposite direction to the DES+DESI $w_0 w_a$ best fit, and of insufficient magnitude.

The background is assumed, not derived. With the tension included, the ejecta are treated as pressureless dust on a flat Λ CDM background once virialized into halos; the $w_{\text{eff}} \approx -0.3$ of the shell toy (Sec. 6.1) describes the shell *kinematics*, not the background equation of state. Under H8 (Precious Rings – Dante’s Purgatory) Geometry A holds inside Kernel 0,0,0 and Geometry B for the rings (Sec. 4). In Geometry A (Sec. 3) the ejection sets the matter’s initial peculiar-velocity distribution on that background, those peculiar velocities redshift into the Hubble flow under the radiation era that relic conversion in Kernel 0,0,0 supplies (Sec. 4), and the supernova, BAO and kSZ comparisons are satisfied because the Λ CDM background is assumed: the ejecta shells of Sec. 3 describe how the matter component acquired its initial velocities rather than an alternative expansion law. Geometry B — the system of rings seen as a whole — has no such shelter. There the expansion law *is* the ballistic ordering of the ejecta, $v = r/t$, plus the tension, and the observational comparison must be made on that law. The ballistic-only LTB lightcone (Appendix C) decelerates ($q_0^{\text{eff}} = +0.16$) and departs from Λ CDM by ~ 0.35 mag by $z \sim 5$; the LTB-plus- Λ lightcone — an inhomogeneous ballistic dust cloud with a constant tension, seen by an interior observer — has *not* been computed. That monopole distance–redshift relation, not the dipole, is Geometry B’s principal observational burden, and until it is computed Geometry B cannot claim the supernova, BAO and kSZ agreement that Geometry A inherits from its assumed background — which, under H8 (Precious Rings – Dante’s Purgatory), is the agreement that matters, since every observation we make is made inside Kernel 0,0,0. What the framework does not supply is the value of the tension: the one framework-native candidate, a residual unconverted vacuum fraction, reproduces the 10^{-123} tuning of standard cosmology (Appendix C), and the origin of ρ_Λ is counted among the open problems (Sec. 13), exactly as in Λ CDM.

7. H2 (Vortex – The Cold Eye of Sauron): The Cold Spot and the Axis of Evil

The Eridanus Cold Spot (Vielva et al. 2004; Cruz et al. 2005) is the sky’s outstanding non-Gaussian feature. On the Planck SMICA map (Planck Collaboration 2020 I, IV, VII) it is centred at galactic $(l, b) = (203^\circ, -56^\circ)$, with a decrement of $-70 \mu\text{K}$ at $5'$ smoothing and a peak of $-168 \mu\text{K}$ at $2'$ smoothing, and a half-maximum radius of $2.5'$; we use these values throughout.

Cyclonic vortex reading. Internal quantum turbulence in the bounce creates macro-vortices. In the **Jovian/cyclonic model** these act as thermal dams that choke local heat transport, deform under shear into elongated features aligned along a preferred rotation axis (the “Axis of Evil”), and — without solid-surface friction to dissipate angular momentum — scale up with expansion and freeze into the CMB at decoupling. The Cold Spot is read as such a frozen cyclone.

Failed prediction: the thermal ring. A cyclone diverting thermal energy around its core would produce a concentric hot ring of $+20$ – $50 \mu\text{K}$ at $5'$ – $7'$. The azimuthally averaged Planck 2018 SMICA profile ($N_{\text{side}} = 128$) around the Cold Spot, with uncertainties from 500 random high-latitude positions (Fig. 5), shows a cold core extending to $\sim 9'$ (-3.4σ at $3'$) and **no hot ring at $5'$ – $7'$** ; a $+37$ – $52 \mu\text{K}$ excess at $13'$ – $15'$ is $\sim 2.5\sigma$ before a trials factor of ~ 10 radial bins. The thermal-ring prediction, the first half of P1 (Polarization Ring), failed. The polarization part survives: a frozen vortex boundary should carry a radial E -mode ring of 1 – $3 \mu\text{K}$ at $\sim 5'$ – $15'$. Planck’s Q_r profile is consistent with zero in every $2'$ ring to $20'$ with per-ring errors of 0.6 – $2.7 \mu\text{K}$ (Appendix A), so the ring is neither detected nor excluded; the test requires CMB-S4 sensitivity.

Preferred axis. Our simulations find ejection isotropic within shot noise (Sec. 3), so an axis can only be supplied by an external seed, the neighbouring bang of Sec. 8. The literature’s cluster of tentative ($\lesssim 3\sigma$) alignments — a supernova acceleration dipole roughly along our CMB motion (Colin et al. 2019; Sah & Rameez 2024), the quadrupole–octupole alignment with the dipole direction,

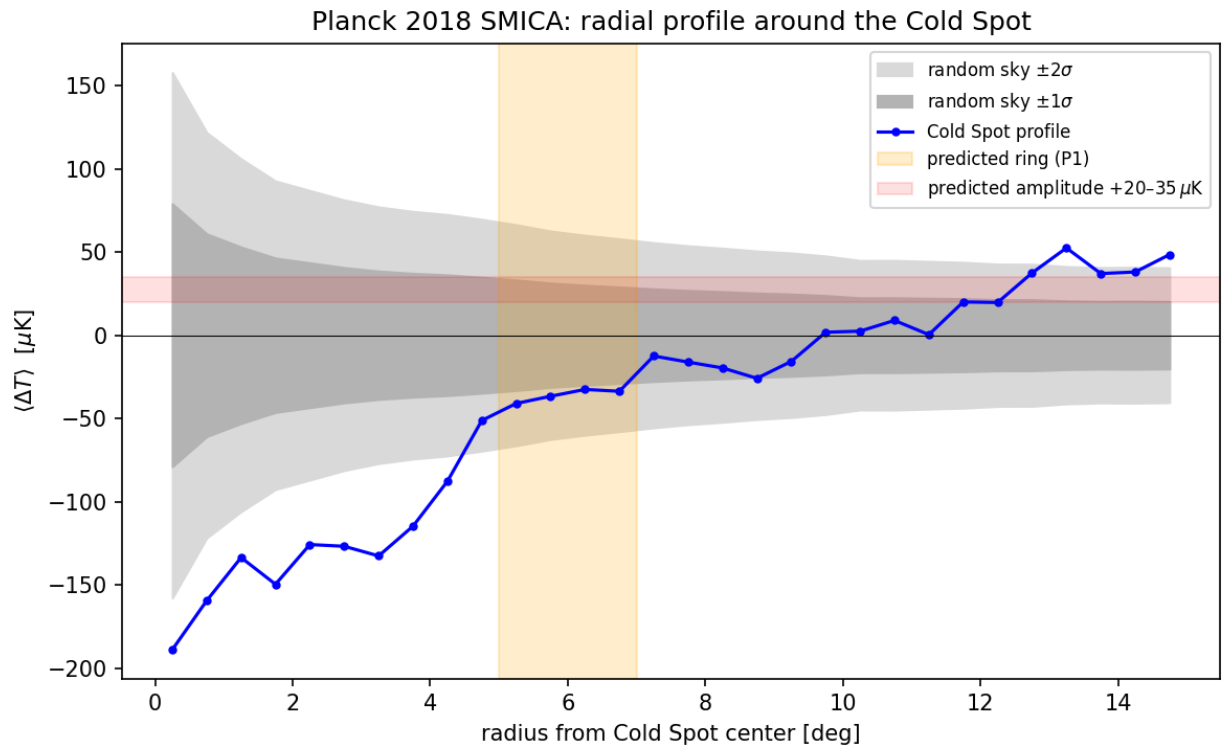


Figure 5: Fig. 5 — Planck 2018 SMICA radial temperature profile around the Cold Spot, vs. 500 random reference positions.

coincident bulk-flow and quasar-polarization axes (Hutsemékers et al. 2005) — is motivation for H2 (Vortex – The Cold Eye of Sauron)/H3 (Neighbour – Big Pool Bang Multiverse 2), not support.

Galaxy handedness. Claimed spin-handedness dipoles (Longo 2011; Shamir 2012, 2022) lie 70° from each other and 50° – 74° from the axes above; we treat them as systematics-limited (Iye et al. 2021). The JWST/JADES handedness claim 8.6° from the Cold Spot is a coincidence awaiting a control-field test (Appendix A).

Falsifier. Absence of the E -mode ring at CMB-S4 sensitivity eliminates H2 (Vortex – The Cold Eye of Sauron). A localized curl detection at the Cold Spot, P8 (Curl Test), would support the vortex reading over the neighbour reading of the same direction (Sec. 8), which is mutually exclusive with it; a calibrated curl null retires it (Table 2).

8. H3 (Neighbour – Big Pool Bang Multiverse 2) and H4 (Population – The Dark Comic Background): A Neighbouring Big Pool Bang in the Cold Spot Direction, and a Population of Neighbours

Because vacuum crystallization is stochastic across an unbounded background, our bang need not be unique. A neighbouring universe nucleating beyond our domain boundary imprints our last-scattering surface, at comoving distance $d_{\text{LS}} \approx 46$ Gly (4.35×10^{26} m); a neighbour closer than that would appear in large-scale-structure tracers, which the Cold Spot sightline does not show beyond the too-shallow Eridanus supervoid.

Causal admissibility. A source at $D = (1 + \epsilon) d_{\text{LS}}$ with $\epsilon > 0$ lies beyond our particle horizon. Nothing emitted by it since the bounce — no black holes crossing the boundary, no gravitational-wave memory, no tidal torque acting over the past Gyr — can have reached our volume. The only causally admissible imprint is one already present in the initial conditions: a potential Φ of the Grishchuk–Zel’dovich type laid down at the bounce, which the last-scattering photons sample as a Sachs–Wolfe spot with a $1/r$ tail over the whole sky at low multipoles. The neighbour reading therefore makes two kinds of prediction: the spot itself (amplitude, profile, non-rotation, polarization edge) and its low- ℓ tail, P9 (Low- ℓ Tail). Its present-day effect on our volume is a static field — an induced (monopole) velocity of the local volume and a tidal shear — fixed once the mass and distance are (Sec. 9).

The Sachs–Wolfe model. For a potential Φ present at last scattering, $\Delta T/T = \alpha \Phi/c^2$ (Sachs & Wolfe 1967), where the coefficient α depends on how the perturbation was laid down: $\alpha = 1/3$ for adiabatic growing modes and $\alpha \rightarrow 2$ for isocurvature-like modes in which the potential is imposed without a compensating density perturbation. An externally imposed potential lies between these limits; we adopt $\alpha = 1$ and carry the resulting factor-of-six range — asymmetric about the adopted value, $M \times 3$ at $\alpha = 1/3$ and $M/2$ at $\alpha = 2$, i.e. $+0.5/-0.3$ dex — into the mass uncertainty of Sec. 9. At the spot centre $\Phi = GM/(\epsilon d_{\text{LS}})$, and the angular profile is

$$\Delta T(\theta) \propto \left[\epsilon^2 + 2(1 + \epsilon)(1 - \cos \theta) \right]^{-1/2},$$

so the half-maximum angle fixes the distance, $\epsilon = \theta_{1/2}/\sqrt{3}$, and the amplitude then fixes the mass, $M = (\Delta T/T) \epsilon d_{\text{LS}} c^2/(\alpha G)$. Because $\theta_{1/2}$ is beam- and smoothing-limited at 2° smoothing, the inferred ϵ values are upper bounds, and since $M \propto \epsilon \Delta T$, the mass at fixed amplitude is an upper bound too. Amplitudes and widths must be paired at the same smoothing; we use the 2° values (ΔT , $\theta_{1/2}$) throughout. The mass function of a population of such spots is the nucleation size spectrum of

the vacuum transition — the distribution that drove the cascade of Sec. 1 — so the internal (ejecta) and external (spot) samples of one process must agree, a cross-consistency requirement linking H1 (Bounce – The Big Pool Bang) and H4 (Population – The Dark Comic Background).

Southern one-sidedness. All eight most-extreme large-scale spots on the SMICA map ($|b| > 25^\circ$, 2° smoothing; Appendix A) lie in the southern galactic hemisphere; in 200 Gaussian realizations through the same pipeline none placed its full top-8 in one hemisphere (mock mean 5.2 of 8). The significance to be quoted is that of the published hemispherical power asymmetry ($p \sim 1\text{--}10\%$, estimator-dependent; Planck Collaboration 2020 VII), since our mocks are simplified and the frame post hoc. What this framework can say is limited: the Cold Spot and the one-sided extremes share a hemisphere, and a neighbour to the galactic south would add a few- μK potential tail on that side of the sky (Sec. 10d). Whether such an additive tail can produce the observed $\sim 7\%$ hemispherical *power* modulation, or additional $\pm 150 \mu\text{K}$ extremes, has not been computed and is not claimed; a few- μK additive term does not obviously do either. The shared hemisphere is therefore a coincidence to be explained, not a prediction of H3 (Neighbour – Big Pool Bang Multiverse 2). An isotropic swarm of neighbours (H4 (Population – The Dark Comic Background)) predicts no such one-sidedness.

H4 (Population – The Dark Comic Background). Since nucleation is stochastic, the neighbour should not be unique: a population of Big Pool Bangs at varying distances, each imprinting a distinct anomaly, P6 (Spot Population). Planck 2018 isotropy analyses (Planck Collaboration 2020 VII) find the sky consistent with Gaussian ΛCDM apart from $2\text{--}3\sigma$ large-scale anomalies and the single Cold Spot, so additional neighbours must imprint below Planck’s per-spot threshold. H4 (Population – The Dark Comic Background) stands or falls on population statistics (Sec. 10), not on any single spot.

Falsifiers. H3 (Neighbour – Big Pool Bang Multiverse 2) falls to a foreground structure accounting for the full decrement, a curl detection at the Cold Spot (which transfers the direction to H2 (Vortex – The Cold Eye of Sauron)), or a P9 (Low- ℓ Tail) null at the pre-registered threshold; H4 (Population – The Dark Comic Background) to a null in sub-threshold spot statistics at LiteBIRD/CMB-S4 depth (Table 2).

9. H6 (Neighbour Location): Where Is the Second Big Pool Bang?

H6 (Neighbour Location) states the neighbour’s location once, summarized in Table 3; other sections refer here.

Direction. The Cold Spot, $(l, b) = (203^\circ, -56^\circ)$, uncertain by $\sim 2\text{--}3^\circ$ (the half-max radius), in the hemisphere of the eight-spot one-sidedness.

Distance. With $\theta_{1/2} = 2.5^\circ$ the profile of Sec. 8 gives $\epsilon = 0.025$; the eight extremes span $\epsilon = 0.025\text{--}0.055$ (Sec. 10). The source lies $\epsilon d_{\text{LS}} \approx 1.1$ Gly comoving beyond the last-scattering surface (up to 2.5 Gly for the widest spots), at $D \approx 47$ Gly $\approx 4.5 \times 10^{26}$ m; since $\theta_{1/2}$ is smoothing-limited, ϵ is an upper bound.

Mass. With $\Delta T = -168 \mu\text{K}$ and $\theta_{1/2} = 2.5^\circ$ (both at 2° smoothing), $T = 2.725$ K, $\epsilon = 0.025$, Sachs–Wolfe coefficient $\alpha = 1$ and $d_{\text{LS}} = 4.35 \times 10^{26}$ m,

$$M_2 = \frac{\Delta T}{T} \frac{\epsilon d_{\text{LS}} c^2}{\alpha G} = (6.2 \times 10^{-5}) (1.09 \times 10^{25} \text{ m}) (1.35 \times 10^{27} \text{ kg m}^{-1}) \approx 9 \times 10^{47} \text{ kg} \approx 10^{48} \text{ kg},$$

uncertain by $+0.6/-0.4$ dex: $+0.5/-0.3$ dex from the Sachs–Wolfe coefficient ($M \times 3$ at $\alpha = 1/3$, $M/2$ at $\alpha = 2$; Sec. 8) and ± 0.3 dex from the range of ϵ across the population (0.025–0.055), added in quadrature. Because ϵ is an upper bound, M_2 at fixed ΔT is an upper bound. The result is $\sim 4 \times 10^{-5}$ of our own per-Hubble-volume $M_1 = 2.2 \times 10^{52}$ kg, a dwarf neighbour. We use $M_2 \approx 10^{48}$ kg ($+0.6/-0.4$ dex) throughout.

Induced velocity of the local volume. The neighbour’s present-day acceleration of our volume is $GM_2/D^2 = (6.67 \times 10^{-11})(9 \times 10^{47})/(4.5 \times 10^{26})^2 \approx 3 \times 10^{-16} \text{ m s}^{-2}$; integrated over a Hubble time ($4.4 \times 10^{17} \text{ s}$) this is $\approx 130 \text{ m s}^{-1}$, the induced (monopole) velocity of the local volume — $\sim 0.1 \text{ km s}^{-1}$, and $\lesssim 1 \text{ km s}^{-1}$ after the mass uncertainty. This is negligible against the $428 \pm 36 \text{ km s}^{-1}$ Cosmicflows-4 bulk flow at $200 h^{-1} \text{ Mpc}$ (Watkins et al. 2023) toward (298ř, $-8ř$), 86ř from the Cold Spot. The neighbour reading therefore predicts *no* bulk-flow contribution above $\sim 1 \text{ km s}^{-1}$, stated as P10 (Induced-Velocity Bound); the observed flow must be explained by local attractors. The genuinely tidal part is the shear: a tidal tensor $GM_2/D^3 \approx 7 \times 10^{-43} \text{ s}^{-2} \approx 1.5 \times 10^{-7} H_0^2$ (with $H_0^2 \approx 4.8 \times 10^{-36} \text{ s}^{-2}$), which integrated over a Hubble time gives a dimensionless expansion-rate anisotropy $\delta H/H_0 \sim 10^{-7}$, undetectable in supernova anisotropy searches, far below the limits on anisotropic expansion (Saadeh et al. 2016) and below the low- ℓ Grishchuk–Zel’dovich (1978) limit — a consistency check, not a discriminant. Hot spots require a boundary imprint of the opposite sign (an underdense or approaching wall); the model does not specify which neighbours imprint hot.

Our position relative to it. $\approx 47 \text{ Gly}$ comoving along the axis toward (203ř, $-56ř$); the internal offset of H5 (Our Position – Where Is Nemo), $\lesssim \text{few Mpc}$ (Sec. 11), is negligible by comparison.

Table 3. H6 (Neighbour Location): the neighbour’s direction, distance, mass and local effects.

Quantity	Estimate	Method	Status
Direction	(203ř, $-56ř$) $\pm 3ř$	Cold Spot centroid; same hemisphere as the eight extremes	Fixed by data; attribution hypothetical
Distance beyond last scattering	$\epsilon = 0.025$, $\approx 1.1 \text{ Gly}$ comoving (upper bound)	$\epsilon = \theta_{1/2}/\sqrt{3}$ at 2ř smoothing	Smoothing-limited
Mass M_2	$\approx 10^{48} \text{ kg}$ ($+0.6/-0.4$ dex; upper bound at fixed ΔT)	Sachs–Wolfe, coefficient $\alpha = 1$ (1/3–2), $\Delta T = -168 \mu\text{K}$ at 2ř	Single adopted value
Induced local velocity (P10 (Induced-Velocity Bound))	$\lesssim 1 \text{ km s}^{-1}$	$GM_2/D^2 \times t_H$ (monopole)	Bound; flow is local
Tidal tensor	$\sim 1.5 \times 10^{-7} H_0^2$ (expansion-rate anisotropy $\delta H/H_0 \sim 10^{-7}$)	GM_2/D^3	Consistent, undetectable
Curl at Spot (P8 (Curl Test))	Zero (neighbour) vs excess (vortex)	Lensing curl / B -mode stack	Uncalibrated null; calibrated test pending

Quantity	Estimate	Method	Status
Low- ℓ tail (P9 (Low- ℓ Tail))	Fixed $\ell = 2, 3$ template	Correlation vs Gaussian null (Sec. 10)	Uninformative at present

What H6 (Neighbour Location) predicts. A *non-rotating* imprint (zero curl at calibrated sensitivity), a disc-shaped edge with the Aguirre & Johnson (2011) bubble-collision polarization profile, a low- ℓ tail of fixed shape (P9 (Low- ℓ Tail)), no bulk-flow contribution (P10 (Induced-Velocity Bound)), and no proper motion. It does not predict the hemispherical power asymmetry (Sec. 8). Falsifiers are in Table 2 (Sec. 12); P8 (Curl Test) is the decisive near-term test, separating the two readings of the same direction.

10. H7 (Pool Model): The Population

(a) Universe 1 is one of several. Vacuum crystallization nucleated more than once; our bang is one of several separated by domain boundaries, each with its own kernel and its own dark rings.

(b) Each neighbour imprints one spot. Under the Sachs–Wolfe model of Sec. 8, the eight most extreme large-scale spots on the SMICA map are the candidate neighbour list (Table 4), each with $\epsilon = \theta_{1/2}/\sqrt{3}$ (an upper bound, since $\theta_{1/2}$ is smoothing-limited at $2\hat{r}$; amplitudes and widths are both the $2\hat{r}$ -smoothing values, and each implied mass carries the $+0.6/-0.4$ dex of Sec. 9):

Table 4. The eight most extreme large-scale spots of the Planck SMICA map ($|b| > 25\hat{r}$, $2\hat{r}$ smoothing).

Rank	(l, b)	Sign	ΔT (μK)	$\theta_{1/2}$	ϵ (upper bound)
1	(157 $\hat{r}$, $-71\hat{r}$)	hot	+170	2.5 $\hat{r}$	0.025
2	(80 $\hat{r}$, $-33\hat{r}$)	cold	-170	3.5 $\hat{r}$	0.035
3	(203 $\hat{r}$, $-56\hat{r}$) (Cold Spot)	cold	-168	2.5 $\hat{r}$	0.025
4	(155 $\hat{r}$, $-29\hat{r}$)	cold	-163	5.5 $\hat{r}$	0.055
5	(305 $\hat{r}$, $-29\hat{r}$)	hot	+153	2.5 $\hat{r}$	0.025
6	(211 $\hat{r}$, $-35\hat{r}$)	cold	-153	2.5 $\hat{r}$	0.025
7	(170 $\hat{r}$, $-47\hat{r}$)	hot	+162	4.5 $\hat{r}$	0.045
8	(184 $\hat{r}$, $-54\hat{r}$)	hot	+155	4.5 $\hat{r}$	0.045

The eight extremes require $\epsilon = 0.025\text{--}0.055$, i.e. sources 1–2.5 Gly (comoving) beyond last scattering. Under Sachs–Wolfe an overdense wall gives a *cold* spot; a hot spot requires an underdensity or an approaching (Doppler) wall, a sign mechanism the model leaves unspecified (Sec. 9). Spot 8 at (184.2 $\hat{r}$, $-54.3\hat{r}$) lies 11 $\hat{r}$ from the Cold Spot centroid (203.2 $\hat{r}$, $-56.3\hat{r}$) and may be part of one system; spot 7 at (170.3 $\hat{r}$, $-46.6\hat{r}$) lies 22 $\hat{r}$ away and is separate. The candidate count is ≤ 8 , or 7 if spot 8 is merged.

(c) Population predictions. Neighbours cluster on the sky (currently 8/8 south). Any genuine neighbour is non-rotating and should show zero curl; any relic vortex of our own bang, H2 (Vortex – The Cold Eye of Sauron), should show a curl excess under P8 (Curl Test) — the population divides on this test. The inferred neighbour mass function must match the nucleation spectrum inferred

from our own ejecta. The exploratory locus, curl, lensing-convergence and tracer statistics on the eight spots are reported once, in Appendix A; none is significant.

(d) The low-multipole tail, P9 (Low- ℓ Tail). A neighbour that imprints a compact spot cannot imprint *only* a spot: the $1/r$ potential of each source extends over the whole sky at $\sim 2\%$ of its peak ($3\text{--}6\ \mu\text{K}$ at 90° , $2\text{--}4\ \mu\text{K}$ at the antipode). Summing the eight tails gives a fixed $\ell = 1\text{--}3$ template with rms $4.1\ \mu\text{K}$ at $\ell = 2$ and $4.5\ \mu\text{K}$ at $\ell = 3$, i.e. $\sim 20\%$ of the observed quadrupole power ($C_2 \approx 210\ \mu\text{K}^2$) and $\sim 8\%$ of the octopole — allowed, and a prediction: the template must correlate positively with the observed $\ell = 2, 3$ maps. *Pre-registered threshold.* With only 5 ($\ell = 2$) and 7 ($\ell = 3$) real modes per multipole, the correlation coefficient r between independent Gaussian patterns has $\sigma \approx 0.4\text{--}0.5$ under the null, so a single value of r is uninformative. The test is: compute r between the template and the masked Planck Commander $\ell \leq 3$ maps (Planck Collaboration 2020 IV), and the same r for 1000 Gaussian ΛCDM realizations (HEALPix synfast, same mask and pipeline); a detection requires r above the 95th percentile of the null at *both* multipoles, and a value at or below the null median at both, with the template amplitude fixed by the spots, falsifies the population reading. An unmasked SMICA comparison ($r = +0.15$ at $\ell = 2$, -0.27 at $\ell = 3$; template axes 78° and 52° from the observed) is uninformative at this power and foreground-contaminated.

(e) One-sided or surrounding. A *surrounding* population would populate both hemispheres; the 8/8 southern count indicates a one-sided, clustered population, and H7 (Pool Model) takes that as its picture. The symmetric lattice of Appendix B is the limiting symmetric case, whose cubic-harmonic ($\ell = 4$) expansion-rate signature is a further falsifier. The pre-registered extension is the hemispheric balance of the extremes ranked 9–28: a one-sided population predicts a persistent southern excess; a surrounding population, or a Gaussian sky with the top-8 count a fluctuation, predicts isotropy. What falsifies H7 (Pool Model) as a whole is given in Table 2 (Sec. 12).

11. H5 (Our Position – Where Is Nemo): Where Are We Relative to Our Own Big Pool Bang?

H5 (Our Position – Where Is Nemo) was conceived as the discriminator between the two geometries of Sec. 3; under H8 (Precious Rings – Dante’s Purgatory) it is a test of where the rings put us. In Geometry A ejection happens everywhere and there is no geometric centre, so H5 (Our Position – Where Is Nemo) predicts a null. In Geometry B the rebound was a centred event and radial ordering survives, so H5 (Our Position – Where Is Nemo) asks: **does any residual large-scale radial ordering survive in our Hubble volume?** Two signatures would answer yes: a displaced-centre dipole ξ in the structure-subtracted peculiar-velocity field above the mock noise floor defined below, or a radial-ordering signature in the bulk-flow profile (an amplitude that grows, rather than decays, with survey depth). Either supports B on our horizon scale; an offset consistent with the mock distribution together with a decaying flow favours A. H8 (Precious Rings – Dante’s Purgatory) predicts the latter: Geometry A inside Kernel 0,0,0, Geometry B for the rings, and we are in Kernel 0,0,0 at or near its centre, so a detected offset above the mock floor would falsify the placement H8 (Precious Rings – Dante’s Purgatory) assigns us. Results are summarized in Table 5.

Displacement from the centre. The machinery is standard (King & Ellis 1973; Alnes & Amarzguoui 2006; Asvesta et al. 2022). Attributing the full CMB dipole ($370\ \text{km s}^{-1}$) to off-centring gives $\Delta r \approx 5\ \text{Mpc}$; the residual unexplained by local gravity gives $\Delta r \lesssim 1\text{--}2\ \text{Mpc}$. A displaced centre is invisible in a purely linear flow and enters only through the curvature of the velocity law, at second order, $\sim H|\xi|^2/2r$: $\approx 90\ \text{km s}^{-1}$ at $r = 10\ \text{Mpc}$ for a $5\ \text{Mpc}$ offset. Fitting the $2\{, \}$ 112 Cosmicflows-4 groups (Tully et al. 2023) at $1.5\text{--}40\ \text{Mpc}$ with $v(R) = hR + kR^2$ diverging from a free centre gives an offset $|\xi| = 6.1\ \text{Mpc}$ toward $(118^\circ, -24^\circ)$ at $\Delta\chi^2 = 6595$, with shells

at 40–80 and 80–150 Mpc returning centres within 10 $\checkmark$ –15 $\checkmark$ of it. Two controls change the picture. *Mock mask*: ten isotropic mocks on CF4’s exact sky positions and distance errors manufacture false centres of $|\xi| = 0.4\text{--}2.5$ Mpc ($\Delta\chi^2 \approx 9\text{--}61$ per shell) that in the outer shells point into the *same longitude band* ($l \approx 93\checkmark\text{--}152\checkmark$) as the data. We define the **mock floor** as the 95th percentile of the structure-subtracted mock $|\xi|$ distribution; with the present ten mocks that percentile is set by the largest false centres, so $|\xi|_{\text{floor}} \approx 2.5$ Mpc, and a tighter value awaits a larger mock ensemble. *Structure subtraction*: removing the 2M++ reconstruction-predicted peculiar velocities (Carrick et al. 2015) drops the inner shell to $\Delta\chi^2 = 182$, $|\xi| = 1.7$ Mpc, within 8 $\checkmark$ of the CMB dipole, and the outer shells into the mock range. The 1.7 Mpc residual lies *inside* the mock distribution (0.4–2.5 Mpc) and is therefore not a detection by the criterion just stated. What makes it worth one more test rather than immediate dismissal is its direction, which is that of the external dipole the 2M++ reconstruction itself calibrates against the CMB frame; whether that direction is signal or calibration is decided by one re-run: repeating the structure-subtracted fit with the 2M++ external dipole left free as a nuisance vector. If $|\xi|$ stays at ~ 1.7 Mpc with the same direction, it is a calibration-independent residual that still sits inside the mock distribution and must wait for the deeper catalogue; if it collapses, it was the calibration. Pending that re-run, **no displaced expansion centre is claimed once mapped structure is subtracted**; the current bound is $|\xi| \lesssim 2$ Mpc, $\sim 10^{-4}$ of the distance to last scattering.

The bulk flow. The bias-corrected Cosmicflows-4 bulk flow (Watkins et al. 2023) is 428 ± 36 km s $^{-1}$ at $200 h^{-1}$ Mpc toward $(298\checkmark \pm 5\checkmark, -8\checkmark \pm 4\checkmark)$, rising with depth, in nominal $\sim 4.8\sigma$ tension with Λ CDM (Whitford et al. 2023 argue the errors are understated). Planck’s kSZ analysis (Planck Collaboration 2014 XIII) limits coherent flows to $\lesssim 254$ km s $^{-1}$ at 95%, but the two numbers refer to different scales: the kSZ limit applies to the coherent motion of clusters at Gpc depth, whereas the CF4 flow is a shallower measurement within $200 h^{-1}$ Mpc, inside which a flow of several hundred km s $^{-1}$ sourced by mapped attractors is not in conflict with it. What the kSZ limit disfavors is a flow of $\gtrsim 250$ km s $^{-1}$ *persisting to Gpc depth*, the second half of P4 (Off-Centre Flow). The flow axis and the CMB dipole ($264\checkmark, +48\checkmark$) lie 63 $\checkmark$ apart, which a single offset vector does not predict; local-attractor infall explains the flow without any offset, and the neighbour of Sec. 9 contributes $\lesssim 1$ km s $^{-1}$. The Cold Spot direction is 86 $\checkmark$ from the flow axis and 116 $\checkmark$ from the CMB dipole, so the southern one-sidedness (Sec. 8) is not an offset signature either.

Table 5. H5 (Our Position – Where Is Nemo): offset and bulk-flow status.

Quantity	Estimate	Method	Status
Offset from centre $ \xi $	$\lesssim 2$ Mpc; 1.7 Mpc structure-subtracted residual, inside the mock distribution	CMB dipole cap; CF4 free-centre fit with mock-mask and 2M++ controls	No detection claimed; free-dipole re-run pending
Mock floor	≈ 2.5 Mpc: 95th percentile of the structure-subtracted mock $ \xi $ distribution (mocks give 0.4–2.5 Mpc)	Ten isotropic mocks on the CF4 mask, $\Delta\chi^2 \approx 9\text{--}61$	Defines the support threshold
Offset direction	Undetermined (CMB dipole ($264\checkmark, +48\checkmark$) vs flow ($298\checkmark, -8\checkmark$))	Dipole/flow attribution	Misaligned by 63 $\checkmark$

Quantity	Estimate	Method	Status
Bulk flow (P4 (Off-Centre Flow))	$428 \pm 36 \text{ km s}^{-1}$ at $200 h^{-1} \text{ Mpc}$; $\gtrsim 250$ km s^{-1} persisting to Gpc depth disfavored	Cosmicflows-4, Planck kSZ	Growth with depth untested beyond $200 h^{-1} \text{ Mpc}$

Support and falsification, in one unit. Both criteria are stated in $|\xi|$ against the mock distribution. *Support for H5 (Geometry B):* a structure-subtracted $|\xi|$ above the 95th-percentile mock floor ($\approx 2.5 \text{ Mpc}$ for the present catalogue), obtained with the 2M++ external dipole left free, with a direction stable across shells; and/or a bulk-flow amplitude that grows with depth beyond $200 h^{-1} \text{ Mpc}$ (P4 (Off-Centre Flow)). *Falsification of Geometry B on our horizon scale:* a structure-subtracted, free-dipole $|\xi|$ consistent with the mock distribution in the next Cosmicflows release, together with a bulk flow that decays with depth. Under H8 (Precious Rings – Dante’s Purgatory) the support criterion and the falsification criterion exchange roles: support for H5 (Our Position – Where Is Nemo) is falsification of the placement of us at the centre of Kernel 0,0,0, and the second outcome is what H8 (Precious Rings – Dante’s Purgatory) requires. The current status is neither: $|\xi| \lesssim 2 \text{ Mpc}$, with the 1.7 Mpc residual inside the mock distribution and the free-dipole re-run pending.²

12. Falsifiable Predictions, Instrumentation, and Falsification Criteria

Hypothesis hierarchy. H1 (Bounce – The Big Pool Bang; P3 (Transition GW Background), P5 (No Particle DM)) is the core and must hold; H8 (Precious Rings – Dante’s Purgatory; P5 (No Particle DM), P11 (Cold Relics)) presupposes H1 (Bounce – The Big Pool Bang) and is the statement that makes its dark matter cold — H1 (Bounce – The Big Pool Bang) can stand without it only as a bounce with hot relics; H2 (Vortex – The Cold Eye of Sauron; P1 (Polarization Ring), P2 (Cyclonic B-modes), P7 (Spin Surcharge), P8 (Curl Test)), H3 (Neighbour – Big Pool Bang Multiverse 2; P8 (Curl Test), P9 (Low- ℓ Tail)), H4 (Population – The Dark Comic Background; P6 (Spot Population)), H5 (Our Position – Where Is Nemo; P4 (Off-Centre Flow)), H6 (Neighbour Location; P8 (Curl Test), P9 (Low- ℓ Tail), P10 (Induced-Velocity Bound)) and H7 (Pool Model; P6 (Spot Population), P9 (Low- ℓ Tail)) are separable from the core: any can fail without falsifying the core mechanism. They are not all mutually independent, however. H2 (Vortex – The Cold Eye of Sauron) and H3 (Neighbour – Big Pool Bang Multiverse 2) are *mutually exclusive* readings of the Cold Spot direction — the Spot is a relic vortex of our own bang or a neighbour’s non-rotating imprint, not both — and the curl test P8 (Curl Test) decides between them: a localized curl excess transfers the direction to H2 (Vortex – The Cold Eye of Sauron), a calibrated null to H3 (Neighbour – Big Pool Bang Multiverse 2). H6 (Neighbour Location) stands or falls with H3 (Neighbour – Big Pool Bang Multiverse 2), and H7 (Pool Model) presupposes H4 (Population – The Dark Comic Background). H8 (Precious Rings – Dante’s Purgatory) and H5 (Our Position – Where Is Nemo) are linked in the opposite sense: a positive H5 (Our Position – Where Is Nemo) result (P4 (Off-Centre Flow)) falsifies the placement of us at the centre of Kernel 0,0,0. The remaining pairs can hold or

²*Status and retirement note (distinct from falsification).* If the next Cosmicflows release, with roughly twice the group count and a $\sim \sqrt{2}$ smaller offset error, returns a structure-subtracted, free-dipole $|\xi|$ consistent with its own (correspondingly tighter) mock distribution, but the bulk flow neither grows nor clearly decays, then H5 (Our Position – Where Is Nemo) has no observable signature at distance-catalogue depth and is *retired as untestable at present*. Retirement is a statement about instrument reach, not a falsification of Geometry B, and is kept out of Table 2’s falsifier column for that reason.

fail independently.

Table 1. Predictions.

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P1 (Polar- ization Ring)	Cold Spot ring (H2 (Vortex – The Cold Eye of Sauron)). Thermal ring of $+20\text{--}50\ \mu\text{K}$ at $5\check{\text{r}}\text{--}7\check{\text{r}}$: failed in Planck 2018 (Sec. 7). Surviving prediction: radial E -mode ring of $1\text{--}3\ \mu\text{K}$ at $5\check{\text{r}}\text{--}15\check{\text{r}}$	Polarization sensitivity $\lesssim 1\ \mu\text{K} \cdot \text{arcmin}$ at $\ell \sim 20\text{--}100$	CMB-S4	Thermal: failed. Polarization: open
P2 (Cyclonic B-modes)	Cyclonic B -mode pattern aligned with the Axis of Evil (H2 (Vortex – The Cold Eye of Sauron))	Degree-scale B -modes, r -equivalent $\sim 10^{-3}$	CMB-S4, LiteBIRD	Open
P3 (Transi- tion GW Background)	First-order- transition stochastic gravitational-wave background from bubble collisions in the crystallization (H1 (Bounce – The Big Pool Bang)): broken power law peaked at the transition scale	Stochastic- background spectral shape at nHz--mHz	Pulsar timing arrays, LISA	Open

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P4 (Off-Centre Flow)	Radial velocity ordering in our Hubble volume — Geometry B’s own prediction (H5 (Our Position – Where Is Nemo)): a structure-subtracted off-centre dipole $ \xi $ above the 95th-percentile mock floor (≈ 2.5 Mpc at present, Sec. 11) with the 2M++ external dipole free and a direction stable across shells, and/or a bulk-flow amplitude that <i>grows</i> with survey depth beyond $200 h^{-1}$ Mpc (currently 428 ± 36 km s $^{-1}$ toward $l \approx 298^\circ$). Geometry A predicts a null: $ \xi $ consistent with the mock distribution and a decaying flow. A flow $\gtrsim 250$ km s $^{-1}$ persisting to Gpc depth is disfavored by the Planck kSZ limit	Next Cosmicflows release (free-dipole re-run); peculiar-velocity survey to $z \sim 1$ with $\lesssim 300$ km s $^{-1}$ per cluster	CF4 (archival re-run); Rubin/LSST + Euclid + kSZ	$ \xi \lesssim 2$ Mpc, 1.7 Mpc residual inside the mock distribution; flow rising to $200 h^{-1}$ Mpc, errors disputed
P5 (No Particle DM)	No particle dark-matter candidate (DM = Planck relics) (H1 (Bounce – The Big Pool Bang))	Continued nulls approaching the neutrino floor	LZ, XENONnT	Ongoing

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P6 (Spot Population)	A sub-threshold population of Cold-Spot-class spots (H4 (Population – The Dark Comic Background); H7 (Pool Model)); Planck already limits any above-threshold population	Sub-threshold spot statistics; per-candidate polarization at $\lesssim 1 \mu\text{K} \cdot \text{arcmin}$; hemispheric balance of extremes 9–28	LiteBIRD, CMB-S4; Planck (archival)	Constrained; open below threshold
P7 (Spin Surcharge)	Spin surcharge (H2 (Vortex – The Cold Eye of Sauron)); filaments and clusters carry primordial angular momentum atop tidal spin, excess largest at high z , randomly oriented (no net spin above shot noise; $\omega/H_0 \lesssim 10^{-9}$; cf. the lattice-spin cap of Sec. 13). The Wang et al. (2021) filament spin is the $z \approx 0$ anchor	Filament-spin stacking at $z \sim 1$ vs $z \sim 0$ against ΛCDM torque mocks	DESI, Euclid, 4MOST	Open

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P8 (Curl Test)	Localized curl excess (lensing curl and B -mode variance) at the extreme-spot positions if they are relic vortices (H2 (Vortex – The Cold Eye of Sauron)); zero at the Cold Spot if it is a neighbour imprint (H3 (Neighbour – Big Pool Bang Multiverse 2); H6 (Neighbour Location)). A calibrated stacked null at the Cold Spot retires the vortex reading (Appendix A)	Planck PR4 curl reconstruction at full depth; degree-scale B -modes at $\lesssim 1 \mu\text{K} \cdot \text{arcmin}$	Planck PR4 (archival), CMB-S4, LiteBIRD	Uncalibrated null; calibrated test pending

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P9 (Low- ℓ Tail)	Low-multipole tail (H3 (Neighbour – Big Pool Bang Multiverse 2); H6 (Neighbour Location); H7 (Pool Model)): the fixed $\ell = 1\text{--}3$ template summed from the eight point-mass profiles (rms $4\,\mu\text{K}$ at $\ell = 2$, $4.5\,\mu\text{K}$ at $\ell = 3$) must correlate positively with the observed $\ell = 2, 3$ maps; detection requires r above the 95th percentile of 1000 Gaussian realizations at both multipoles (Sec. 10d)	Masked Commander low- ℓ maps (Planck Collaboration 2020 IV); r against the simulated null	Planck 2018/PR4 (archival)	Unmasked $r = +0.15$ ($\ell = 2$), -0.27 ($\ell = 3$): uninformative
P10 (Induced- Velocity Bound)	Induced-velocity bound (H6 (Neighbour Location)): the neighbour's induced (monopole) velocity of the local volume is $\lesssim 1$ km s^{-1} (Sec. 9), so it contributes nothing measurable to the bulk flow; the observed $428\,\text{km s}^{-1}$ must be accounted for by local attractors	Bulk-flow residual after 2M++/CF4 reconstruction $\ll 400\,\text{km s}^{-1}$ unexplained	CF4 (archival); DESI peculiar velocities	Bound; consistent

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P11 (Cold Relics)	Cold relics (H8 (Precious Rings – Dante’s Purgatory)): the dark-matter thermal velocity dispersion at equality is below $10^{-6} c$ for any conversion temperature above the MeV scale ($p/m_{PC} = \gamma/R \approx \gamma T_{\text{eq}}/T_{\text{conv}}$, Sec. 4), i.e. zero free-streaming cutoff in the matter power spectrum down to the smallest scales: no warm-dark-matter suppression	Small-scale matter power: no cutoff at any k ; equivalently $m_{\text{WDM}} \rightarrow \infty$ in warm-dark-matter parametrizations	Lyman- α forest, strong-lens substructure, 21-cm	Consistent ($m_{\text{WDM}} \gtrsim \text{few keV}$); not yet discriminating

Table 2. Falsification criteria. This is the single reference for what each hypothesis needs and what removes it; Secs. 10, 13 and the Epilogue refer here.

Hypothesis	What supports it	What falsifies it	Current status
H1 (Bounce – The Big Pool Bang) bounce and ejection	$88\% \pm 4\%$ ejected fraction at $\sigma_{\text{rel}} \approx 1$; particle-DM nulls (P5 (No Particle DM)); a transition-shaped GW background (P3 (Transition GW Background))	A particle dark-matter detection; an ejecta velocity spectrum that cannot yield a cold matter component by matter–radiation equality; once a derivation of ρ_c from the jamming limit exists, a derived value differing from the LQC $0.41 \rho_P$ by more than a factor of 3	Open; relativistic formulation unresolved; ejecta temperature resolved by location under H8 (Precious Rings – Dante’s Purgatory), pending the freeze-out calculation

Hypothesis	What supports it	What falsifies it	Current status
H8 (Precious Rings – Dante’s Purgatory) rings: Kernel 0,0,0 with cold relic survivors, dark rings outside	Displaced-centre null (offset inside the mock distribution) with a decaying bulk flow; no free-streaming cutoff in the small-scale matter power (P11 (Cold Relics)); particle-DM nulls (P5 (No Particle DM)); a merge-then-evaporate freeze-out that reproduces $f = T_{\text{eq}}/T_{\text{conv}}$	Any of: a detected off-centre dipole above the mock floor with a direction stable across shells (we would not be at the centre of Kernel 0,0,0); a measured dark-matter thermal velocity dispersion above γ/R (a keV-scale warm-dark-matter detection); failure of the freeze-out calculation (merger rate, remnant stability) to reproduce $f = T_{\text{eq}}/T_{\text{conv}}$; any evidence that the CMB region has an edge within the horizon	Open; displaced-centre null and Lyman- α limits consistent; freeze-out calculation not done
H2 (Vortex – The Cold Eye of Sauron) Cold Spot vortex	E -mode ring (P1 (Polarization Ring)), cyclonic B -modes (P2 (Cyclonic B-modes)), curl excess at the Spot (P8 (Curl Test)), spin surcharge (P7 (Spin Surcharge))	Absence of the E -mode ring at CMB-S4 sensitivity; a calibrated stacked curl null at the Cold Spot	Thermal ring failed; polarization open; curl uncalibrated
H3 (Neighbour – Big Pool Bang Multiverse 2) neighbour in the Cold Spot direction	Zero curl at the Spot (P8 (Curl Test)); positive low- ℓ tail correlation (P9 (Low- ℓ Tail)); bubble-collision polarization edge	A foreground structure accounting for the full decrement; a curl detection at the Spot; P9 (Low- ℓ Tail) at or below the null median at both multipoles	Open; P9 (Low- ℓ Tail) uninformative
H4 (Population – The Dark Comic Background) population of neighbours	Sub-threshold spot population (P6 (Spot Population)); mass function matching the ejecta nucleation spectrum	Null sub-threshold spot statistics at LiteBIRD/CMB-S4 depth	Constrained above threshold; open below

Hypothesis	What supports it	What falsifies it	Current status
H5 (Our Position – Where Is Nemo) residual radial ordering (Geometry B vs A)	Structure-subtracted, free-dipole $ \xi $ above the 95th-percentile mock floor (≈ 2.5 Mpc at present) with a direction stable across shells; bulk flow growing with depth (P4 (Off-Centre Flow)). Either supports Geometry B	Structure-subtracted, free-dipole $ \xi $ consistent with the mock distribution in the next Cosmicflows release, together with a bulk flow decaying with depth: falsifies Geometry B (retirement as untestable is a separate status, Sec. 11 footnote)	Unresolved; $ \xi \lesssim 2$ Mpc, 1.7 Mpc residual inside the mock distribution, awaiting the free-dipole re-run
H6 (Neighbour Location) location of the second bang	Non-rotating imprint (P8 (Curl Test)); low- ℓ tail (P9 (Low- ℓ Tail)); no induced-velocity contribution to the bulk flow (P10 (Induced-Velocity Bound)); Aguirre–Johnson polarization edge	Curl detection at the Spot; absence of the bubble-collision polarization profile at CMB-S4 sensitivity	Open
H7 (Pool Model) one-sided population	Southern excess persisting in extremes 9–28 (P6 (Spot Population)); positive P9 (Low- ℓ Tail); zero curl at all eight spots	All of: P9 (Low- ℓ Tail) failing at the pre-registered threshold; zero curl at calibrated sensitivity at all eight spots <i>and</i> no bubble-collision polarization profile at any at CMB-S4 depth; no CMB-independent mass counterpart in a pre-registered galaxy-density re-run. For the lattice limit: a detected $\ell = 4$ cubic expansion-rate pattern (Appendix B)	Open; 8/8 south at $p \sim 1\text{--}10\%$

Under the joint nulls of H3 (Neighbour – Big Pool Bang Multiverse 2), H4 (Population – The Dark Comic Background), H6 (Neighbour Location) and H7 (Pool Model) the spot population is Gaussian and the Cold Spot a single outlier of unknown origin, while H1 (Bounce – The Big Pool Bang), H8

(Precious Rings – Dante’s Purgatory) and H2 (Vortex – The Cold Eye of Sauron) stand and H5 (Our Position – Where Is Nemo) continues to test whether we sit at the centre of Kernel 0,0,0 as H8 (Precious Rings – Dante’s Purgatory) requires. The polarization ring and calibrated curl test await Simons Observatory / CMB-S4 (a 2030s timeline); P9 (Low- ℓ Tail) and P10 (Induced-Velocity Bound) are archival.

13. Limitations and Open Problems

1. **Hard-sphere ansatz.** Treating Planck relics as incompressible spheres is a modeling assumption motivated by the Compton–Schwarzschild coincidence, not a quantum-gravity result.
2. **Bounce mechanism and horizon structure.** The rebound requires the LQC-form modified Friedmann equation (Bojowald 2001; Ashtekar & Singh 2011); $\rho_c = \eta_{\text{rcp}}\rho_P$ is a conjecture. The bounce is meaningful without qualification only for a whole or super-horizon patch (Geometry A). The finite-cloud Geometry B *currently has no bounce mechanism*: a finite cloud of the relevant mass is $\sim 10^{40}$ inside its Schwarzschild radius, the homogeneous LQC equation offers it no exit, and no trapped-surface timing analysis — which would also have to accommodate the multi-bubble assembly timescale ($\gtrsim 10^{20} t_P$ per Hubble volume) — has been done. Geometry B is retained only as the geometry H5 (Our Position – Where Is Nemo) can test kinematically.
3. **Spectral index.** ν_t and the relation $n_s - 1 = -\nu_t/3$ are fitted inputs; the compatibility of a Kolmogorov cascade with near scale invariance is unproven.
4. **Energy budget: jamming bounce versus cold relics.** The $88\% \pm 4\%$ ejection requires $\sigma_{\text{rel}} \approx 1$, which the cascade model reaches only marginally, and the end-to-end chain is untested. The bounce-to-equality budget (Sec. 3) shows that if a single homogeneous population of relics sources the radiation era its ejecta are still relativistic at equality ($p/m_{PC} \approx 1.7$ for any γ), i.e. hot dark matter. H8 (Precious Rings – Dante’s Purgatory; Sec. 4) resolves the conflict by location — the relics that fell back into Kernel 0,0,0 are ground into light and a cold trace of survivors, the ejecta are the dark rings outside — but at a price that is stated, not paid: (a) the merge-then-evaporate freeze-out calculation (merger rate, remnant stability, the conversion temperature T_{conv}) that must reproduce the survivor fraction $f = T_{\text{eq}}/T_{\text{conv}}$ has not been done; (b) baryogenesis in the bath is not supplied by the channel, so the local dark-matter-to-baryon ratio of 5.4 is not derived; (c) the claim that Kernel 0,0,0 is homogeneous on our horizon scale — that we see no edge and no centre — is an assumption tested only by the displaced-centre null. Planck-relic dark matter must also face femtolensing, capture, and abundance constraints, and the relativistic formulation of the ejection remains open.
5. **Expansion history.** Neither the Newtonian toy nor the LTB lightcone produces apparent acceleration, the toy’s $w(z)$ drift runs opposite to the DES+DESI direction, and the kSZ gate caps any ejecta contribution at $\sim 1\%$ of dark energy. The adopted history (Sec. 6.2) is the Big Pool Bang framework plus a constant tension on an assumed flat Λ CDM background (Geometry A); the tension’s value is not predicted, and Geometry B’s own monopole distance–redshift relation (the LTB-plus-tension lightcone) has not been computed. The model is also not yet reconciled with BBN abundances or the CMB acoustic peaks.
6. **Causal and relativistic structure.** At the bounce the causal patch is of Planck

size and one present-day Hubble volume comprises $\sim 10^{60}$ of them; homogeneity across causal patches must be assumed (the horizon problem stands), and in Geometry A, where the jammed patch is super-horizon, its total extent is unconstrained. Ejection proceeds where Newtonian escape velocity vastly exceeds c ; the N -body mechanism lacks a relativistic formulation, prerequisite for H1 (Bounce – The Big Pool Bang) to be a physical theory rather than an analogy. H8 (Precious Rings – Dante’s Purgatory) assigns the two geometries of Sec. 3 by location — Geometry A inside Kernel 0,0,0, Geometry B for the rings — and H5 (Our Position – Where Is Nemo) tests that assignment; whether the boundary between Kernel 0,0,0 and the mantle is sharp, and where it lies relative to our horizon, is not modelled. The neighbour of H3 (Neighbour – Big Pool Bang Multiverse 2)/H6 (Neighbour Location) lies beyond the particle horizon, so its imprint must be an initial-condition potential (Sec. 8); how such a potential is laid down across a domain boundary is not modelled.

7. **Relic-to-plasma conversion.** Kernel 0,0,0 must convert relics to Standard-Model plasma with the observed baryon asymmetry while ejected relics stay stable. A merge-then-evaporate channel is conjectured in Sec. 3 (merger products above the remnant mass evaporate; isolated relics do not) and made quantitative in H8 (Precious Rings – Dante’s Purgatory; Sec. 4) as a freeze-out with survivor fraction $f = T_{\text{eq}}/T_{\text{conv}}$; it is untested in the Planck regime, the conversion temperature is not derived, and it supplies no baryon asymmetry.
8. **Sign of hot spots and the Sachs–Wolfe coefficient.** Sachs–Wolfe supplies cold spots from overdense walls; the hot extremes require an unspecified underdensity or Doppler mechanism (Sec. 9). The coefficient relating Φ to $\Delta T/T$ for an externally imposed potential is bracketed (1/3–2, a factor of six) rather than derived, which is the dominant term in the +0.6/–0.4 dex on M_2 .
9. **Flatness and one-sidedness.** Spatial flatness is not addressed by the energy budget of the rebound (Sec. 5). Whether a neighbour’s few- μK potential tail can generate the observed hemispherical power modulation has not been computed (Sec. 8).

Each item maps onto the observational program of Sec. 12 (Tables 1 and 2); the theory is offered as a falsifiable mechanical alternative, not a completed replacement for ΛCDM .

Further research: spin and an equal-mass lattice. A bang born from an isotropic bounce carries no net spin above shot noise (the per-sky ejection dipole of Sec. 3 matches the shot-noise expectation); like a planet or a galaxy it acquires spin only through tidal torque from neighbours, at second order (rate $\sim q GM/D^3$ for a cloud of asymmetry q). If every lattice bang is fixed at our own mass per Hubble volume, $M_1 = 2.2 \times 10^{52}$ kg, two order-of-magnitude constraints follow that we have not developed here. First, the $1/r$ potential tail of an equal-mass neighbour keeps the CMB quadrupole below $\sim 4 \mu\text{K}$ only for spacings $D \gtrsim 10^3$ Gly ($\sim 20 d_{\text{LS}}$); at the Cold Spot distance an equal-mass neighbour would imprint a ~ 4 K spot, so under Geometry B the Cold Spot source is either a bang of order 10^{-5} – 10^{-4} of our mass or not a bang at all. In Geometry B the cloud mass is at least the matter mass of the observable universe, $\gtrsim 1.5 \times 10^{53}$ kg, which raises both numbers: the equal-mass spot becomes $\gtrsim 28$ K and the quadrupole-safe spacing $D \gtrsim 2 \times 10^3$ Gly. Second, because $\omega \propto q GM/D^3$ and the quadrupole bound already fixes the minimum spacing, the tidally induced global spin of an equal-mass lattice is *capped*, at $\omega/H_0 \lesssim 10^{-8}$ ($q/0.1$) with shear $\sim 10^{-7}$. A spin detection therefore cannot calibrate the lattice from above: any detected global spin above that cap — at $10^{-6} H_0$, say — would *contradict* the equal-mass lattice rather than calibrate it, whereas a detection at or below

the cap would fix the spacing. The relevant existing bound is that of Saadeh et al. (2016), who limit the vorticity of homogeneous Bianchi VII_h modes to $(\omega/H_0)_0 \lesssim 10^{-10}$ from the Planck temperature and polarization quadrupole and spiral patterns. Whether that bound applies here is an open question that we state honestly rather than settle: it is derived for homogeneous *decaying* vorticity modes, whereas a tidally induced spin is a growing second-order mode; but its CMB signature is the same quadrupole/spiral pattern already bounded, and if the Saadeh bound transfers directly the equal-mass spacing rises to $D \gtrsim 5 \times 10^3$ Gly. Late-time velocity-field curl statistics reach, as an order-of-magnitude estimate, only $\omega/H_0 \sim 10^{-2}$ and add nothing. The connection to P7 (Spin Surcharge) is the $\omega/H_0 \lesssim 10^{-9}$ quoted there: the primordial spin surcharge of P7 (Spin Surcharge) is randomly oriented and averages to zero on large scales, whereas the lattice spin is a single global axis, and the two are separately bounded. A treatment of tidal-torque spin-up of a finite cloud in Geometry B, with the shear and spin axes and their CMB signatures, is left to a separate paper.

Two roads not taken. (i) *Domain walls as dark energy.* The wall between our bang and a neighbour has tension, and a network of such walls does accelerate expansion, but with $w = -2/3$, excluded by the measured $w = -1.03 \pm 0.03$; a single wall beyond the horizon exerts no force on the interior at all (Sec. 6.2, Appendix B). We therefore keep the constant tension and do not attribute it to walls. (ii) *A pre-bounce contraction.* Bouncing cosmologies usually solve the horizon problem with a long contracting phase before the bounce. We do not adopt one: the framework starts from a false vacuum that crystallizes, not from a previous universe, and a contracting pre-history would move the initial-condition question one cycle back while adding its own problems (growth of anisotropy and curvature during collapse, entropy carried across the bounce). We prefer to carry the horizon problem openly (Sec. 5) rather than import a second origin story to close it.

Data and code availability. Code, figures, build instructions and download links for all datasets used are at <https://github.com/AIMFIRST-VN/big001>.

Epilogue: Three Questions, Three Answers

We set out with three questions and a pool table. Here is where the balls came to rest.

How does it work? Picture the moment before the first moment: a false vacuum, tense as a drawn bow. Somewhere a bubble of true vacuum nucleates, its wall sweeping outward, catalysing more bubbles, until the whole patch is a jammed crush of Planck-mass spheres pressed to the random-close-packing limit — some 10^{60} of them in what will become each Hubble volume, and no end to the patch that we can see. Nothing can fall further. A loop-quantum bounce fires, the pile rebounds, and the rebound is chaotic: $88\% \pm 4\%$ of the spheres are hurled outward, sorted by speed like shot from a gun. What happens next depends on where each sphere lands, and that is H8 (Precious Rings – Dante’s Purgatory): one kind of ball, one number for each — its kick against the escape speed — and a different law on every ring. The balls that fell back make the innermost ring, Kernel 0,0,0, and there they collide: two Planck relics that merge are heavier than a relic is allowed to be, and the merger boils away into light, over and over, until the ring has thinned so far that the mergers stop — a freeze-out, exactly as for a WIMP — and what is left is a bath of light with a cold trace of surviving spheres in it. That bath is the hot Big Bang: the microwave background, the helium, the baryons. Those survivors are our dark matter. No inflaton, no reheating, no residual vacuum: the light came from the relics. The balls that rose and fell back late make a dark mantle, mixed by many orbits, that remembers nothing. The 88% that escaped are the dark rings around us — cold, dark, sorted by speed, the only ring that still keeps the geometry and spin of the bang, and the one we will never see except as a fossil in the initial conditions. In Kernel 0,0,0 the crush was everywhere and had no centre: Geometry A, in which the ejection set the matter’s initial peculiar

velocities on a homogeneous background and those velocities redshift into the Hubble flow under the radiation era the grinding in Kernel 0,0,0 supplies. Seen as a whole, the rings are one centred event: Geometry B, in which the fastest went furthest and that ordering *is* a ballistic Hubble-like law, $v = r/t$, to which the tension below is added — at the price of having, as yet, no bounce mechanism for a finite cloud (which sits 10^{40} inside its own horizon, and nothing in the equations brings it out) and of a monopole distance–redshift relation that has not yet been computed. Either way the debris does not push the universe faster; nothing outside a volume ever can, and nothing inside it pushes unless it has negative pressure. The acceleration we measure comes from a constant tension of space, $w = -1$, which we adopt on an assumed Λ CDM background and cannot yet derive. So the model works as far as the matter goes, hands the expansion to a tension it does not explain, and carries three open problems: the freeze-out that fixes how many relics survive in Kernel 0,0,0 has been written as one relation, $f = T_{\text{eq}}/T_{\text{conv}}$, and not yet computed, and the baryons it must also make are not accounted for; the thermal ring around the Cold Spot is not there; and the bounce is only meaningful without qualification if the jammed patch was the whole universe rather than a ball inside its own horizon.

Where are we? In Kernel 0,0,0, says H8 (Precious Rings – Dante’s Purgatory) — the innermost ring, the only one with light, at or near its centre — and this is the question that tests it. Inside Kernel 0,0,0 the rebound happened everywhere, there is no centre to be far from, and the test must come up empty; that is Geometry A, and it is what the rings require of us. If instead we sat off-centre in a centred cloud, the signature would be P4 (Off-Centre Flow): an off-centre dipole ξ in the structure-subtracted galaxy velocity field above the mock noise floor — the 95th percentile of the false centres that isotropic mocks manufacture on the same sky mask, about 2.5 Mpc at present — or a bulk flow that grows rather than fades with depth. We looked, and after the mock-mask and 2M++ controls the answer so far is: no offset claimed, two megaparsecs at most, with a 1.7 Mpc residual that lies inside the mock distribution and points along the CMB dipole, which one re-run — the 2M++ external dipole left free — will either confirm as calibration-independent or dissolve as a calibration. What we do have is a bulk flow of $428 \pm 36 \text{ km s}^{-1}$ at $200 h^{-1}$ Mpc, which the kinetic Sunyaev–Zel’dovich data do not forbid at that depth but would forbid if $\gtrsim 250 \text{ km s}^{-1}$ of it persisted to a gigaparsec, and a sky whose extreme spots all lie in the southern hemisphere. The reading we adopt is that we are not visibly off-centre, exactly as a position at the centre of Kernel 0,0,0 requires, and that the lopsidedness of the sky is more plausibly the work of a neighbour than of our own position. Geometry A inside Kernel 0,0,0, Geometry B for the rings: the theory now says which is which, and the measurements of Table 2 can say whether it is right, since a detected offset above the mock floor would put us somewhere H8 (Precious Rings – Dante’s Purgatory) says we are not. Where we are remains the least answered of the three questions, and the one whose answer would settle the most.

Where is the second Big Bang? Follow the coldest patch of the oldest light: galactic longitude 203°, latitude -56° , the Cold Spot. If it is the fingerprint of a neighbouring Big Pool Bang written into the initial conditions of our last-scattering surface, then that neighbour sits just beyond it, about one billion light-years past the edge of everything we can see, with a mass of order 10^{48} kg , uncertain by a factor of four upward and two and a half downward — a dwarf beside the $2 \times 10^{52} \text{ kg}$ in each of our own Hubble volumes. It lies beyond our horizon, so nothing it has done since can reach us: no black holes, no gravitational-wave echoes, and an induced drift of our neighbourhood of a tenth of a kilometre per second, too small to move anything we can measure. Seven other extreme spots lie on the same side of the sky, a candidate pool of neighbours clustered rather than surrounding us — though whether a neighbour can lopsided the sky’s power, rather than merely share its hemisphere, is not something we have computed. The neighbour reading makes promises it can

be held to. It must leave no curl-mode fingerprint at the spot, where a vortex would — P8 (Curl Test), which decides between the two readings because the Spot cannot be both. Its $1/r$ potential must leak a few microkelvin across the whole sky in a pattern fixed by the spots — P9 (Low- ℓ Tail), to be judged against a thousand Gaussian skies rather than by eye. And it must contribute nothing to the bulk flow — P10 (Induced-Velocity Bound), which it does not. The neighbour hypothesis will not survive as a story; it will survive, or fall, as the set of numbers in Table 2.

What surprised us

Six things we did not expect when we racked the balls, and one thing we were glad to lose.

A one-sided sky. All eight of the most extreme spots in the Planck map lie in the southern galactic hemisphere, none in the north, and the Cold Spot is one of them.

A spiral in the Cold Spot. The Cold Spot has the most coherent spiral winding of any of the 384 sky positions we scanned, a slow 40° twist from centre to rim: what a frozen vortex would look like, and what a Gaussian fluctuation has no reason to do.

Neighbours push, but not outward on average. Surrounding bangs do pull on our galaxies: harder along the lines toward them, and inward in the gaps between them. Gauss’s law makes the sky-averaged pull exactly zero at every radius, and the directional part is about 10^{-7} of the Hubble rate. The only thing that can push isotropically is mass inside our own volume, and the kinetic Sunyaev–Zel’dovich gate caps what our own ejecta can contribute at about 1% of the observed acceleration. That is why the paper had to hand the acceleration to a constant tension of space.

Equal-mass neighbours must be very far away. If every bang had our own mass, a neighbour at the Cold Spot distance would burn a 4 K hole in the microwave sky instead of a $168\ \mu\text{K}$ dimple. So either the Cold Spot source is a bang some 10^5 times lighter than ours or it is something else entirely, and equal-mass bangs must keep a spacing of at least a thousand billion light-years — several thousand if the CMB vorticity bound applies to a tidally induced spin — so that any global spin they induce is capped near 10^{-8} of the Hubble rate, and a larger one would rule the equal-mass lattice out.

The neighbour is beyond the horizon. If it exists, nothing it has done since the bounce has reached us. Everything we can test is a fossil potential laid down before the first light.

The rings solve the budget. We spent a long time trying to make one history do two jobs — a jammed rebound that ejects 88% of the mass, and a radiation era that leaves cold relics — and the energy budget refused every version of it. The answer was not a new ingredient but a map: the same relics, different physics by ring. The ones that fell back are ground into light and a cold trace in Kernel 0,0,0; the ones that escaped are the dark rings outside. We live in Kernel 0,0,0, the innermost ring and the only one with light, which is why there is anyone here to notice.

What we got rid of: the dark-matter particle, and the inflaton

The largest simplification the framework buys is the one it was built for. Dark matter here is not a new particle. It is the Planck-mass relics that survived the merger freeze-out in Kernel 0,0,0 — the cold trace left when the grinding that made the light stopped keeping pace with the expansion (H8 (Precious Rings – Dante’s Purgatory)) — ordinary gravitating mass with no charge, no light, and no reason to appear in any detector but a gravitational one; the $88\% \pm 4\%$ that the rebound threw outward is the same stuff, but it is the dark rings outside Kernel 0,0,0, not the dark matter in our galaxies. Forty years of direct searches, from xenon tanks to axion cavities, have found nothing,

and in this picture that is the expected result rather than an embarrassment: P5 (No Particle DM) turns the null into a prediction, and a confirmed particle detection would falsify H1 (Bounce – The Big Pool Bang) outright. What the framework does *not* get rid of is dark energy. The constant tension of space is adopted, its value is not derived, and Section 6.2 says so. One of the two dark sectors is explained by mechanics; the other is inherited. The second thing we lost is comic inflation. There is no inflaton, no potential to tune, no slow roll and no eternal-inflation multiverse of bubbles: the universe does not inflate, it bounces, and the bounce is what lets it start without a singularity and without the Borde–Guth–Vilenkin past boundary. The price is stated as plainly as the gain. Inflation was invented to make the sky flat and uniform, and nothing in this framework yet does those two jobs; the horizon and flatness problems are carried as open problems, and the red tilt of the primordial spectrum is a fitted number, not a derived one. So the scorecard reads: dark-matter particle gone, inflaton gone, dark energy kept, two of inflation’s chores still waiting for a mechanism, and the jamming bounce and cold dark matter reconciled by rings — with the freeze-out that the reconciliation rests on still to be computed. That is the honest scorecard.

When we will know more

Most of the decisive tests are archival and cheap. The curl search (P8 (Curl Test)) at calibrated sensitivity and the low-multipole template (P9 (Low- ℓ Tail)) can be run on Planck PR4 products within a year; a null on the first retires the vortex reading of the Cold Spot, a null on the second retires the population reading of the extremes. The JADES handedness control field is a single reprocessing of public JWST data. Our own position (P4 (Off-Centre Flow)) waits on the next CosmicFlows release and on DESI and Euclid peculiar velocities, roughly 2027–2029, which should tighten the mock floor by $\sim \sqrt{2}$ and settle whether the 1.7 Mpc residual is calibration — and, with it, whether we sit at the centre of Kernel 0,0,0 as H8 (Precious Rings – Dante’s Purgatory) says. The cold-relic prediction, P11 (Cold Relics), is already being tested by every Lyman- α and lens-substructure limit on warm dark matter, all of which it passes; it becomes decisive only if one of them finds a cutoff. The polarization ring and cyclonic B -modes, P1 (Polarization Ring) and P2 (Cyclonic B -modes), need degree-scale polarization at the microkelvin-arcminute level: Simons Observatory results toward 2028, CMB-S4 and LiteBIRD in the early 2030s. The gravitational-wave background from a first-order transition, P3 (Transition GW Background), is a LISA-era question, after 2035. By the end of this decade, then, the Cold Spot will be a vortex, a neighbour, or neither, and we will know whether we sit visibly off-centre in our own Big Pool Bang.

In one paragraph: a jammed crystal that bounced, a speed-sorted debris field arranged in rings — the innermost, Kernel 0,0,0, ground into the light we live in and the cold trace we call dark matter; the outer ones dark, and the seed of the Hubble flow or the Hubble flow itself depending on which ring one asks — and a cold fingerprint on the far wall that may belong to someone else. Each of these is stated so that a measurement can remove it.

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Appendix A. Exploratory Null Results and a Pre-registered Test

This appendix records the exploratory battery run on the eight most extreme large-scale spots of the Planck SMICA map ($N_{\text{side}} = 128$, $|b| > 25^\circ$, 2° smoothing; per-spot amplitude ΔT and half-maximum radius θ ; positions in Sec. 10). Nulls are 60–300 Gaussian realizations drawn from the map’s own pseudo- C_ℓ and passed through the identical pipeline unless stated otherwise.

Locus test. If neighbours shared a common distance shell and a common formation density, their spots would follow a locus $|\Delta T| \propto \theta^3$, whereas Gaussian peaks obey BBKS statistics (Bardeen et al. 1986) with amplitude and size nearly independent. A log–log fit of $|\Delta T|$ against θ gives slope -0.00 , $r = -0.01$, 27th percentile of the Gaussian null ($+0.07 \pm 0.10$). No amplitude–size correlation is detected, but the test is powerless as implemented (radii quantized to 2.5° – 5.5° , amplitude spread $\sim 10\%$). It is reported as unresolved pending injection-recovery validation, not as an exclusion. Neighbours at a range of distances would populate a diffuse cloud rather than a locus, and one genuine neighbour among seven Gaussian impostors would not move the fit.

Internal dipoles and spin budget. Monopole-plus-dipole fits within 6° in the local tangent frame: the eight extremes split 4/4 hot/cold with rank-matched amplitudes summing to $-11 \mu\text{K}$ against a mean $|\Delta T| \approx 162 \mu\text{K}$ (7% of Gaussian skies as balanced); opposite-sign spots lie 4.0° closer than same-sign (null $+5.2^\circ \pm 16.3^\circ$, $p = 0.30$); nearest-neighbour spacing is no more regular than Gaussian. Converting dipoles to spin-axis proxies ($\mathbf{L} \propto \hat{n} \times \mathbf{d}$), the Cold Spot’s internal dipole ($16.4 \mu\text{K}/\text{deg}$) is the largest by a factor 2.3 and the others weakly co-align. These fits are dominated by $\ell \sim 2$ –20 gradients that Gaussian peaks generically carry and that are correlated across co-hemispheric spots, so they are upper bounds on rotation, not spin measurements; a gradient-subtracted re-fit is required before any spin fraction can be quoted.

Spiral winding. The azimuthal phase of the $m = 1$ mode in 1.5° rings from 1.5° to 15° around the Cold Spot is locked ring-to-ring with coherence 0.988 — above all 300 random positions (null 0.72 ± 0.18) — and drifts by $\sim 3^\circ$ of phase per degree of radius ($\sim 40^\circ$ total twist). A static

background gradient gives high coherence but zero winding; only the winding is spin-specific. The $m = 2, 3$ modes are unremarkable, and this is a post-hoc test.

Curl search. The public PR4 lensing release (Carron, Mirmelstein & Lewis 2022) contains only gradient modes, so we implemented an approximate temperature-only quadratic estimator ($N_{\text{side}} = 256$, $\ell \leq 512$, spin-1 curl projection, mean field from 40 Gaussian simulations; unnormalized, valid only as a relative comparison). Stacking curl-potential variance in 6 r discs: the eight spots sit at the 82nd percentile of 60 simulated skies and the Cold Spot at the **38th** — indistinguishable from a random position. This does not trigger the retirement clause of P8 (Curl Test), which is bound to a sensitivity reaching the temperature-dipole-implied rotation that this reduced-resolution estimator does not attain (Planck’s full-depth reconstruction uses $\ell \leq 2048$ with polarization).

Polarization. On the SMICA Q/U maps ($N_{\text{side}} = 256$, $\ell \leq 512$, E/B decomposition, 1 r smoothing): stacked B -mode variance in 6 r discs at the eight spots is at the 66th percentile of random positions, E -mode variance at the 35th, and the Cold Spot’s radial- E profile is consistent with zero in every 2 r ring to 20 r with per-ring errors of 0.6–2.7 μK — the predicted 1–3 μK ring of P1 (Polarization Ring) is neither detected nor excluded.

Lensing convergence. The sign-weighted PR4 κ stack (Planck Collaboration 2020 VIII; Carron et al. 2022) on the eight positions is +0.006 against a random-position null of 0.000 ± 0.013 (4/8 sign matches): no integrated mass scars. The Cold Spot’s inner-2 r $\kappa = -0.070$ vs null $+0.013 \pm 0.042$ (-2σ) has the sign of its known shallow foreground supervoid; the 8 r –20 r annulus never exceeds $+0.6\sigma$, so there is no mass “belt”. A stack on $\sim 27,000$ ordinary small spots (1 r smoothing, $|T| > 1.5\sigma$) shows cold spots on κ deficits (-7.2σ) and hot on excesses ($+2.2\sigma$), but κ is reconstructed from the same temperature data and quadratic estimators leak temperature extrema into the mass map; whether this is leakage or genuine ISW $\times$ lensing is undiscriminated, and a polarization-only κ stack is the required discriminating check.

CMB-independent tracers. Against the Quiaia Gaia–unWISE quasar catalogue (Storey-Fisher et al. 2024; 756k quasars, $z \sim 1\text{--}3$, selection-function corrected) the small-spot correlation vanishes (cold -1.1σ , hot -0.3σ) and the eight extremes are within $\sim 1\sigma$ of random (Cold Spot -0.018 , 0.6σ); this is not a sensitive null, since Quiaia’s kernel barely overlaps the ISW kernel. Against WISE $\times$ SuperCOSMOS (Bilicki et al. 2016; 17.7M galaxies, $z = 0.1\text{--}0.4$, high-passed at 10 r): small spots remain null (cold -0.7σ , hot $+0.2\sigma$), while the extremes show seven of eight sign matches and a sign-weighted stack of $\approx +2.7\sigma$ against a correlated-noise null from 300 random 5 r discs. Qualifications: the result rests on three sightlines ($|z| \approx 2.0\text{--}2.3$); the nominal binomial 3.5% ignores the three-tracer, multi-statistic search (global $p \sim 10\text{--}30\%$); extremeness-conditioning inflates the recovered correlation (Eddington-type bias); the 10 r high-pass self-subtracts genuine $\gtrsim 10\text{r}$ supervoids; and extinction/stellar residuals at $|b| = 25\text{--}40\text{r}$ are of order the per-spot signals. Verdict: a 1–2 σ -class hint that the extremes are partly sourced by $z \sim 0.1\text{--}0.4$ superstructures, to be confirmed or retired by a pre-registered re-run at two baseline scales with $|b| > 30\text{r}$ and extinction regression. The 2M++ cross-match of the eight sightlines ($< 180 h^{-1} \text{ Mpc}$) matches spot signs at coin-flip rates (5 of 8).

JWST handedness. The JADES field at (224 r , -54r), 8.6 r from the Cold Spot, shows a claimed 2:1 handedness asymmetry in 263 high-redshift galaxies (Shamir 2025). The sample is small, the method contested, and a Doppler boost from the Milky Way’s rotation is a mundane candidate. The discriminating measurement is a control field — the identical pipeline on COSMOS or GOODS-N ($> 100\text{r}$ away): a systematic reproduces the asymmetry everywhere; a Cold Spot spin column produces it only on the column. The local anchor is already measured: the same-handedness angular

correlation over 77{,}840 SDSS galaxies (11.2M pairs, 0.05ř–4ř) is zero in every bin, with a 1.9σ global excess.

Coherence-gradient hypothesis (adopted by H8 (Precious Rings – Dante’s Purgatory)).

In the N -body toy, only the earliest, fastest ejecta escape in a single pass and preserve geometric correlations; everything slower undergoes violent relaxation, which conserves spin magnitude but randomizes spin geometry. If any residual radial ordering survived in our Hubble volume — H5 (Our Position – Where Is Nemo) — one would expect a *coherence gradient*: fossil geometry intact in the earliest-ejected material, degrading toward the well-mixed later ejecta. Intermediate tracers (quasar-polarization alignments at $z \sim 1$ –2; galaxy handedness axes) lie 24ř–73ř from the Cold Spot direction, consistent with random axes, and the local same-handedness correlation above is zero to a few parts in 10^3 , as mixing requires. H8 (Precious Rings – Dante’s Purgatory; Sec. 4) makes this its statement — free streaming preserves initial geometry and spin in the shell, collisions in Kernel 0,0,0 and orbit mixing in the mantle erase them — with the consequence that the coherent material is outside our horizon and the material we can survey is the mixed material; the local nulls are what it predicts. It has no validated observable of its own inside Kernel 0,0,0.

Statistical accounting. Across the battery (locus slope, hemisphere count, hot/cold balance, pairing distance, lattice regularity, dipole magnitudes, cancellation ratio, winding coherence, κ stack, curl percentile, tracer stacks) the individual results are $p \sim 0.06$ –0.7; under any global look-elsewhere correction the sub- 2σ items are jointly consistent with noise. Further method limits: the Gaussian nulls use the unmasked pseudo- C_ℓ without mask deconvolution, anisotropic noise, or foreground residuals (end-to-end NPIPE simulations are the required upgrade for any sub-percent claim), and the spots were selected on the same map on which all statistics were evaluated, so the hypotheses were partly formulated post hoc. What survives is qualitative: the extreme-spot sky is one-sided at literature-level significance, the Cold Spot is the outlier in every channel examined, and nothing requires a population of additional objects.

Pre-registered commitment (P8 (Curl Test)). The load-bearing, falsifiable claim of the vortex reading is the curl test. We commit in advance that a clean stacked curl-mode null at the Cold Spot position, at a sensitivity reaching the temperature-dipole-implied rotation, *retires* the vortex interpretation of that direction in favor of the non-rotating neighbour reading (H3 (Neighbour – Big Pool Bang Multiverse 2)/H6 (Neighbour Location)) — not merely disfavors it. If curl variance is instead detected localized there, the vortex reading stands and the Cold Spot becomes a relic vortex of our own bang rather than a neighbour’s imprint.

Appendix B. A Fixed Grid of Surrounding Bangs (Consistency Check)

The population model in its most symmetric form — the limiting case of the one-sided population H7 (Pool Model) adopts (Sec. 10e) — places Big Pool Bangs on a fixed lattice with our own at one site. As a consistency check, we ask whether the neighbours’ pull on our ejecta could contribute to the observed late-time acceleration. Point masses of equal mass were placed on a cubic grid of unit spacing (26 neighbours for $3 \times 3 \times 3$, 124 for $5 \times 5 \times 5$); test galaxies were placed at distance r from us in random directions and the vector sum of the neighbours’ inverse-square pulls was evaluated (Table 6). Units: the pull of one neighbour on us.

Table 6. Net pull of a cubic lattice of equal-mass neighbours on test galaxies at distance r from us.

r (grid units)	Sky-averaged outward pull	Toward a neighbour (axis)	Toward a gap (diagonal)
0.1	0.0000	+0.012	−0.008
0.2	0.000	+0.10	−0.06
0.3	0.000	+0.34	−0.20
0.4	0.00	+0.85	−0.44
0.45	0.00	+1.26	−0.60

The results (Table 6) are identical for both grid sizes. Galaxies lying toward a neighbour are pulled outward with a pull that grows steeply with r ; galaxies lying toward the gaps between neighbours are pulled inward; the sky average is zero at every radius. This is Gauss’s law, not a property of the arrangement: the sky-averaged radial acceleration at radius r is $GM_{\text{enc}}(r)/r^2$, and external sources enclose nothing. Only mass, or negative pressure, *inside* the observed volume can change the isotropic expansion rate. The lattice therefore predicts (i) zero contribution to the monopole acceleration, and (ii) a direction-dependent expansion rate with the cubic-harmonic ($\ell = 4$) pattern of the lattice, fast toward the six nearest neighbours and slow toward the eight diagonal gaps. Observation (ii) is a further falsifier for the lattice limit of H7 (Table 2): supernova and BAO expansion-rate anisotropy is limited to the few-percent level with no cubic pattern, while the amplitude implied by the neighbour mass and distance of Sec. 9 is $\sim 10^{-7}$ at the $r \approx 0.3$ reached by the supernova sample. The one form in which surrounding mass does raise the local expansion rate isotropically is a local underdensity, where the effect comes from the mass *missing inside*. A $\delta \approx -0.3$ underdensity within ~ 300 Mpc, as claimed by Keenan, Barger & Cowie (2013), would imply an outflow of ~ 1000 km s $^{-1}$ at its edge, in tension with the kSZ gate of Sec. 6; the literature is divided (Keenan–Barger–Cowie 2013 for; Wu & Huterer 2017 and Kenworthy, Scolnic & Riess 2019 against), and this paper takes no position on whether the void exists. What it does conclude is that such a void cannot be the acceleration: a Gpc-scale void deep enough to mimic Λ is excluded by the kSZ gate. This appendix is a consistency check on the adopted expansion history of Sec. 6.2, not its basis.

Appendix C. Kinematic Mimicry of Acceleration: Tests

This appendix records the tests summarized in Sec. 6 of whether the ejecta distribution could mimic acceleration for an interior observer.

Relativistic (LTB) recalculation. Dust shells in the Lemaître–Tolman–Bondi metric obey the same radial equation as Newtonian shells; what the toy of Sec. 6.1 gets wrong is the *observables*. Integrating radial null geodesics through the evolving LTB metric (300 shells seeded from the ejecta speed distribution, fallback mass virialized into the interior, $d_L = (1+z)^2 R$) gives $q_0^{\text{eff}} = +0.16$ ($w_{\text{eff}}(0) \approx -0.23$): mild deceleration, no apparent acceleration, with a distance-modulus shape ~ 0.35 mag from both Λ CDM and coasting out to $z \sim 5$. Scope restrictions: the lightcone threads only the free-streaming 9%, the interior treatment and observer epoch are modelling choices, and shell crossing was not monitored.

Profile inversion. Since the ejection profile is an input, we searched power-law, bimodal fast-tail, and stochastically lumpy profiles for any that accelerates. Smooth families robustly decelerate ($q_0^{\text{eff}} \sim +1$ to $+4$); apparently negative values in lumpy configurations are seed-ensemble noise (± 1.4 , exceeding the target $|q_0| \approx 0.55$, so that family is unconstrained rather than excluded). The known way an LTB model mimics acceleration — the giant void, observer inside a Gpc-scale underdensity — is the *opposite* density ordering from outward ejection.

Blast wave and the kSZ gate. A rebound acting as a blast wave would sweep ejecta into a shell and leave a rarefied cavity, the giant-void geometry; swept-shell profiles do give smoother lightcones (0.03–0.08 mag residuals vs ~ 1 mag). This is not a rescue: the comparison metric cannot separate Λ CDM from coasting (~ 0.04 mag rms apart), a 0.08 mag residual corresponds to $\Delta\chi^2 \sim 400$ against Pantheon+, a cavity deep enough for $q_0 \approx -0.55$ is the giant-void model excluded by kinetic Sunyaev–Zel’dovich and Compton- y constraints (Zhang & Stebbins 2011), and mass swept into a distant shell is unavailable as local dark matter. Imposing the constraints first: the kSZ limit on coherent cluster flows ($v \lesssim 300$ km s $^{-1}$ at Gpc distances) caps the mean interior contrast of any cavity at $|\bar{\delta}| \lesssim 0.05$ (500 Mpc) to $\lesssim 0.01$ (3 Gpc), i.e. $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. No ejection profile of any morphology supplies more than $\sim 1\%$ of the observed dark energy; this is the result quoted in Sec. 6.

A framework-native candidate, with its cost. If crystallization is incomplete today, a residual unconverted vacuum fraction would be smooth, non-clumping, and immune to the kSZ gate — qualitatively dark energy. Quantitatively this re-imports the cosmological-constant problem verbatim ($\rho_P \sim 10^{96}$ vs 6×10^{-27} kg m $^{-3}$: tuning to one part in 10^{123}), and ongoing conversion would imply present-epoch nucleation, domain walls (the Zel’dovich, Kobzarev & Okun 1974 bound), and an unconverted component at BBN and recombination — none computed here. The boiling picture does anchor H3 (Neighbour – Big Pool Bang Multiverse 2)–H4 (Population – The Dark Comic Background) in Coleman & De Luccia (1980) bubble nucleation, whose collisions imprint disc-shaped CMB features with specific polarization profiles (Aguirre & Johnson 2011) and, generically, source the stochastic gravitational-wave background of P3 (Transition GW Background).